

Discrete Geometry: Binary Images

- Images consist of only two colors
 - E.g. white & black

0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	255	255	255	255	0	0
0	0	255	255	255	255	0	0
0	0	255	255	255	255	0	0
0	0	255	255	255	255	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

- Image is characterized by location of white (or black) pixels

Matrix Representation:

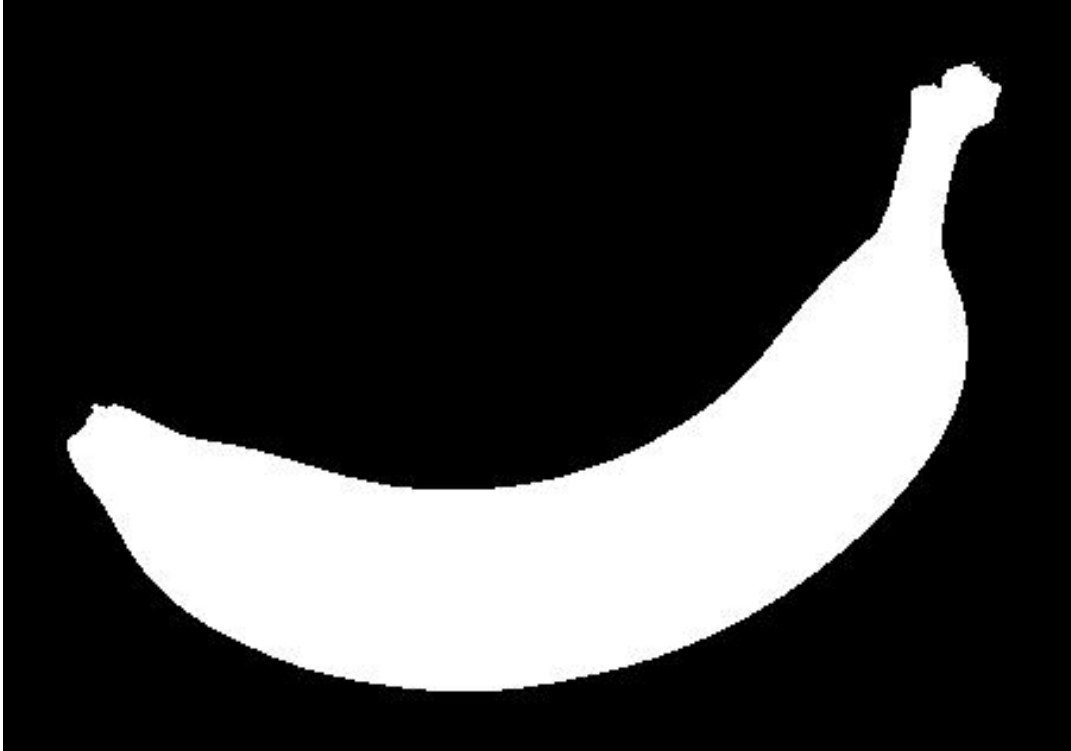
0	0	0	0
0	0	0	0
0	0	1	1
0	0	1	1

Set Representation:

Locations of white (or black) pixels:

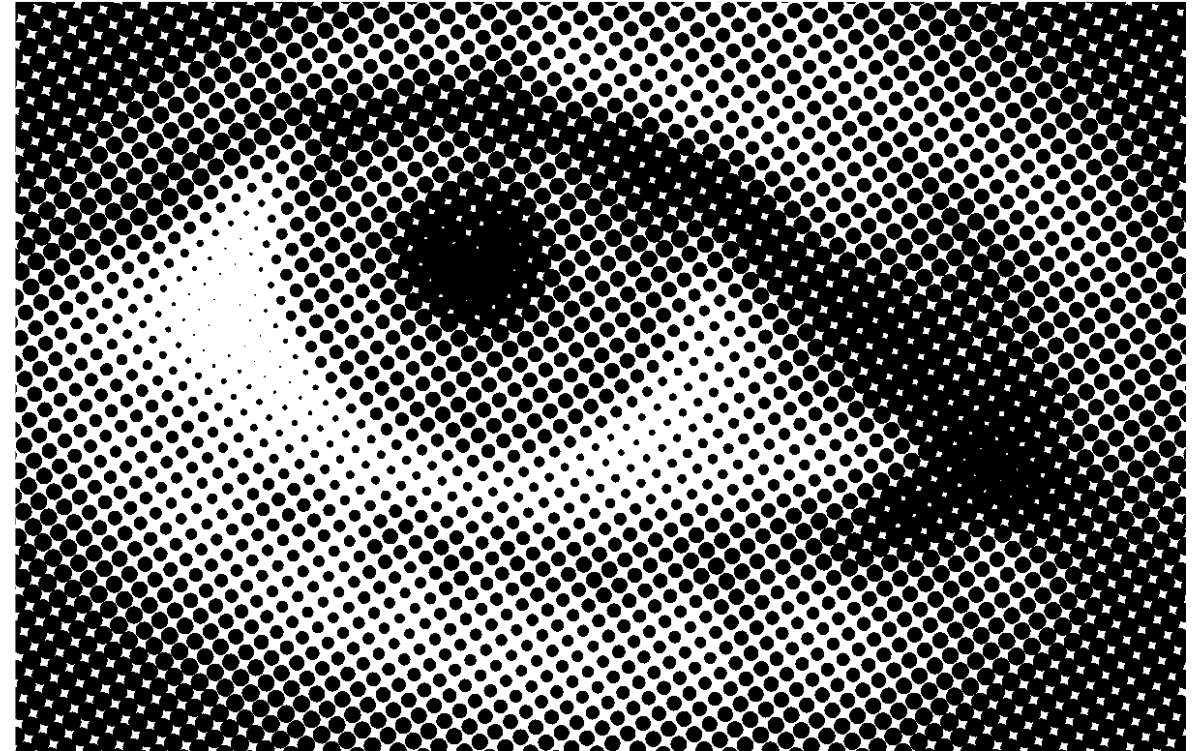
$L=\{(3,3),(3,4),(4,3),(4,4)\}$

Binary Images



Segmentation

Each region different color

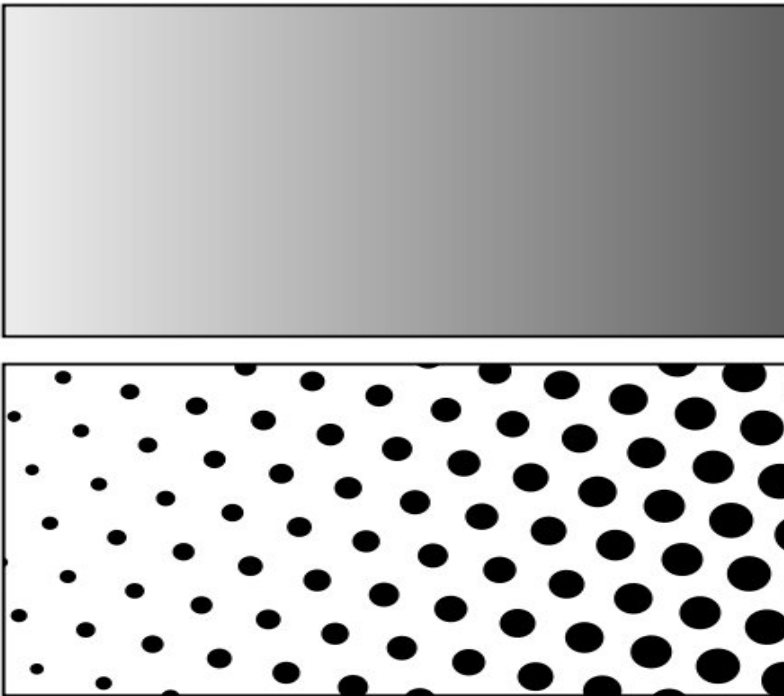


Printing: Half-Tone

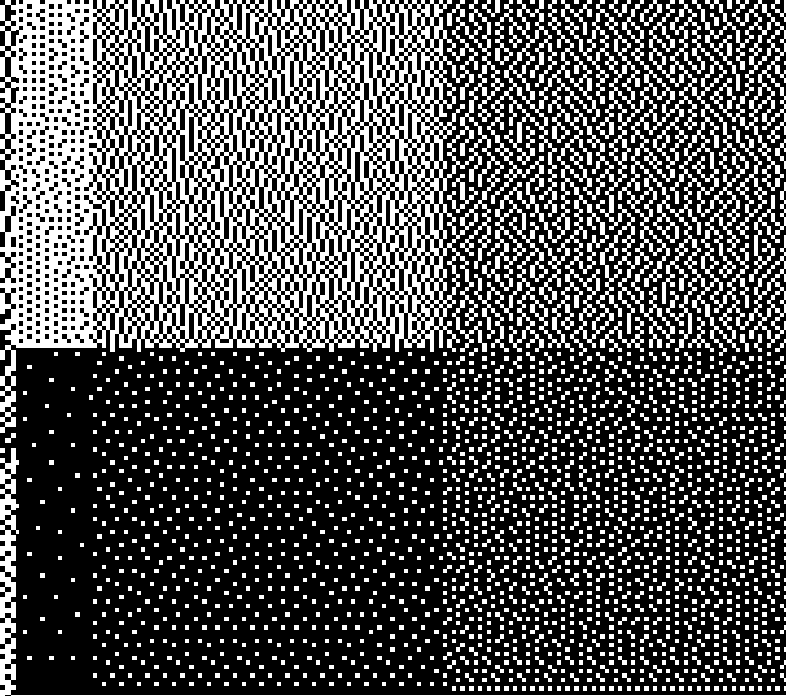
Represent grey with B/W
Ink does not have grey level...

Halftoning: Screen Offset Printing

Dither Inkjet Printing

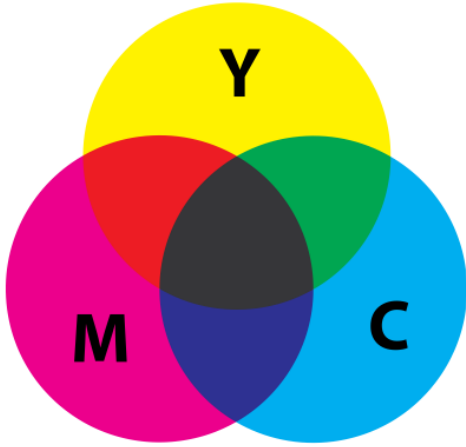


Screen (Books, Offset):
Variable size dots on a
fixed grid.
Intensity - Dot size



Dither (Inkjet):
fixed size dots in random
locations. Intensity - Dot
distribution

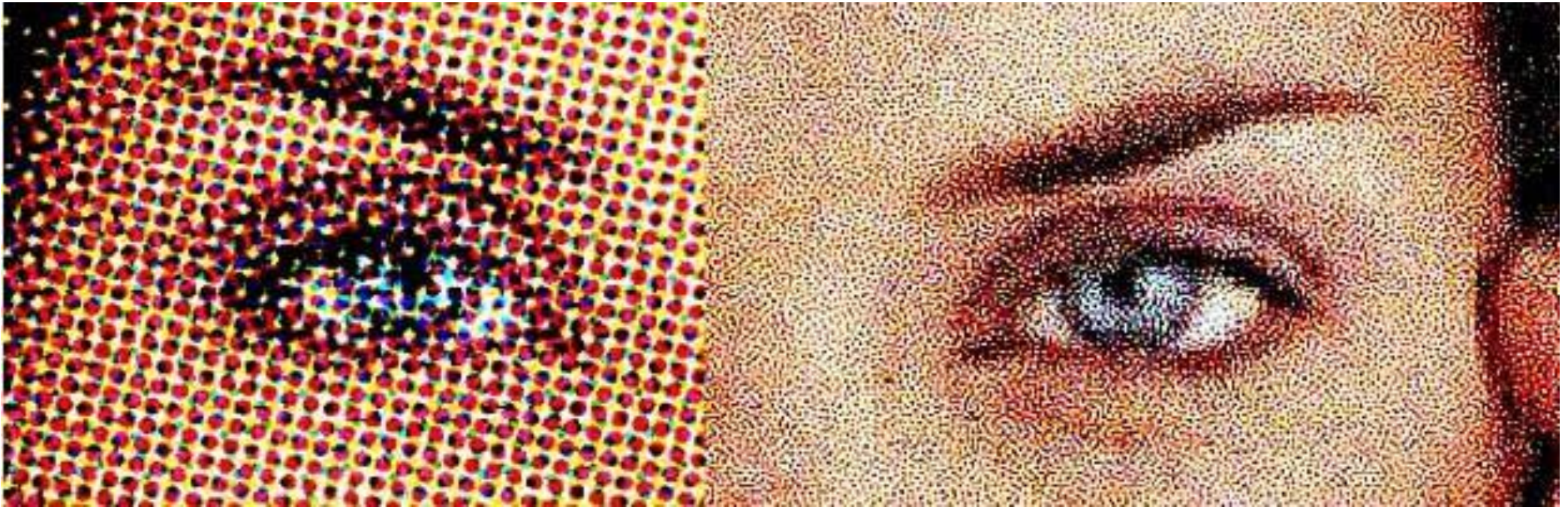
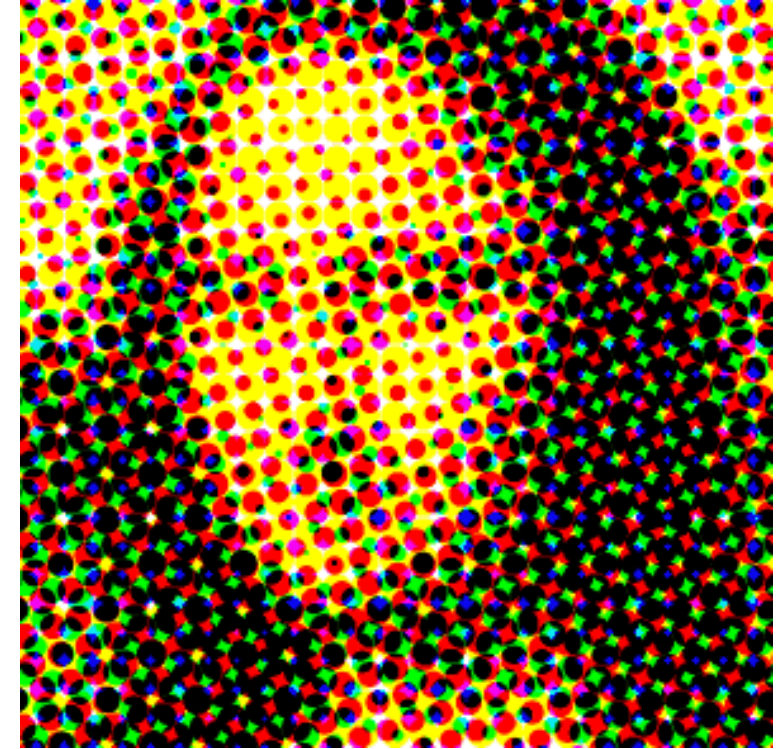
Printing: Color Half-Toning



Color Printing

4 screens: CMYK

Problem: Moire between
color screens

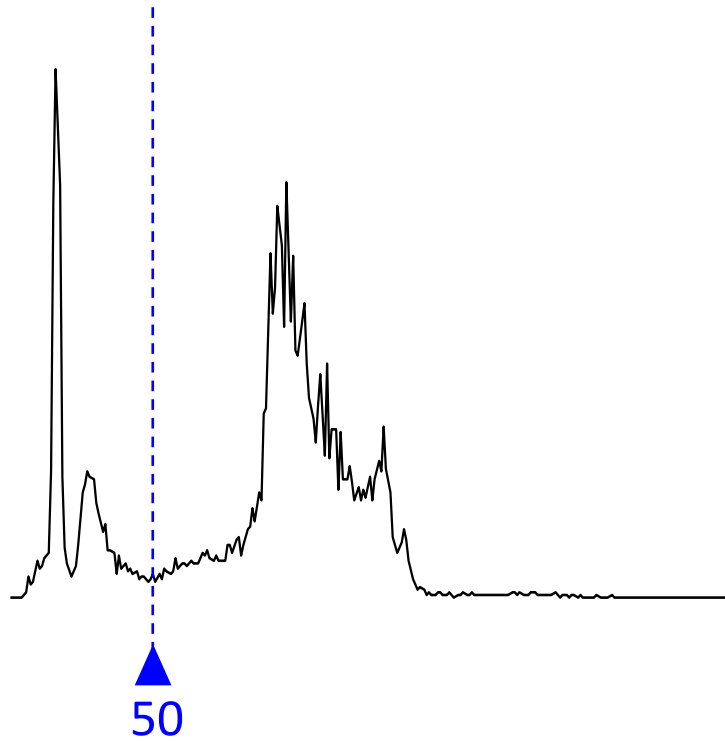


Binarization by Thresholding

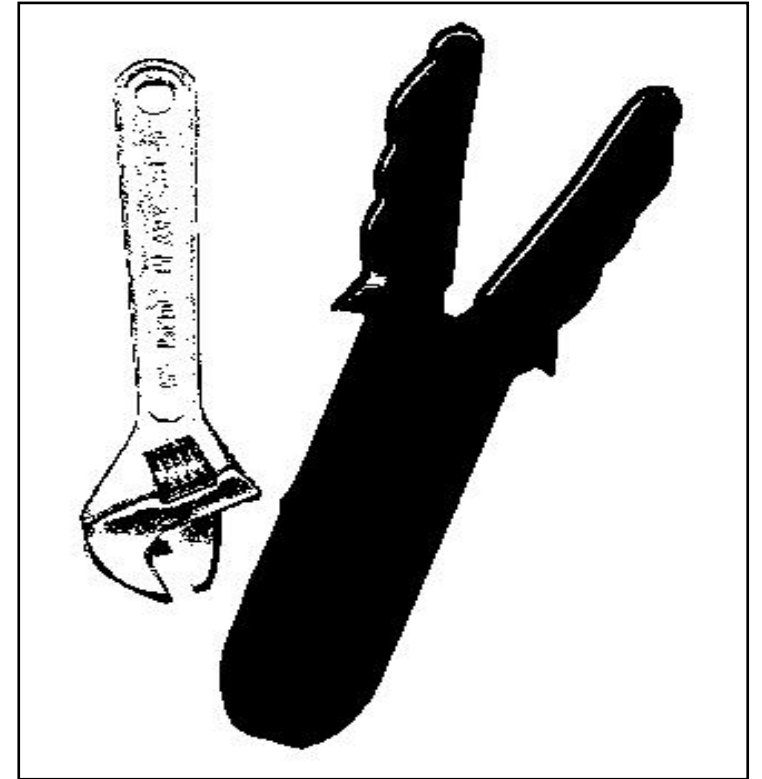
Original



Histogram



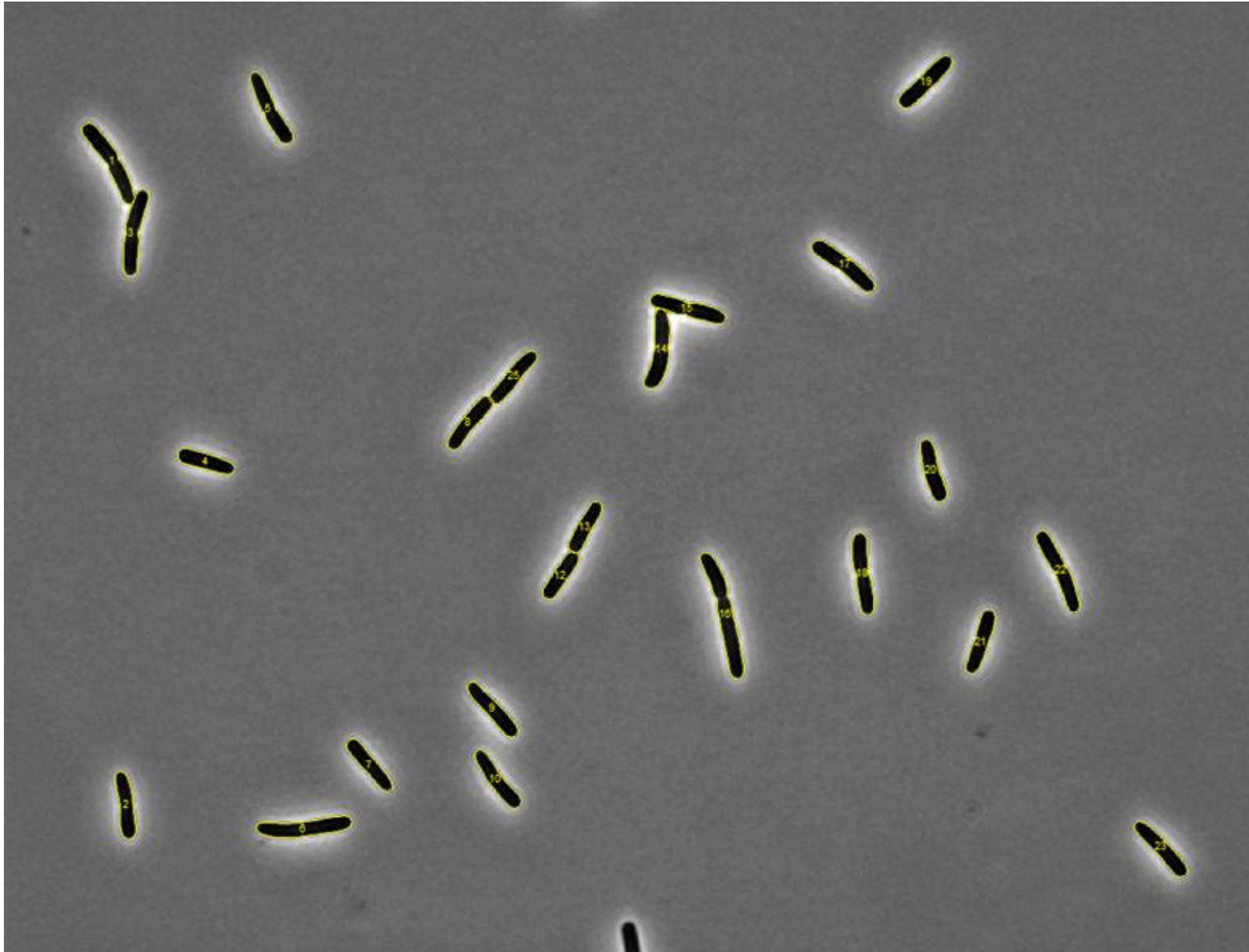
Threshold = 50



Threshold Selection

- Valley in the histogram
 - A little problematic. How deep? How wide?
- Use the gradient, since object boundary has high gradients
 - Compute image gradient (E.g. Sobel)
 - Compute the average grey level at pixels with high gradient
 - Use this average as threshold
- Threshold Scanning
 - Scan through several thresholds
 - Select the threshold whose boundary has highest gradient levels.

Bacteria Segmentation

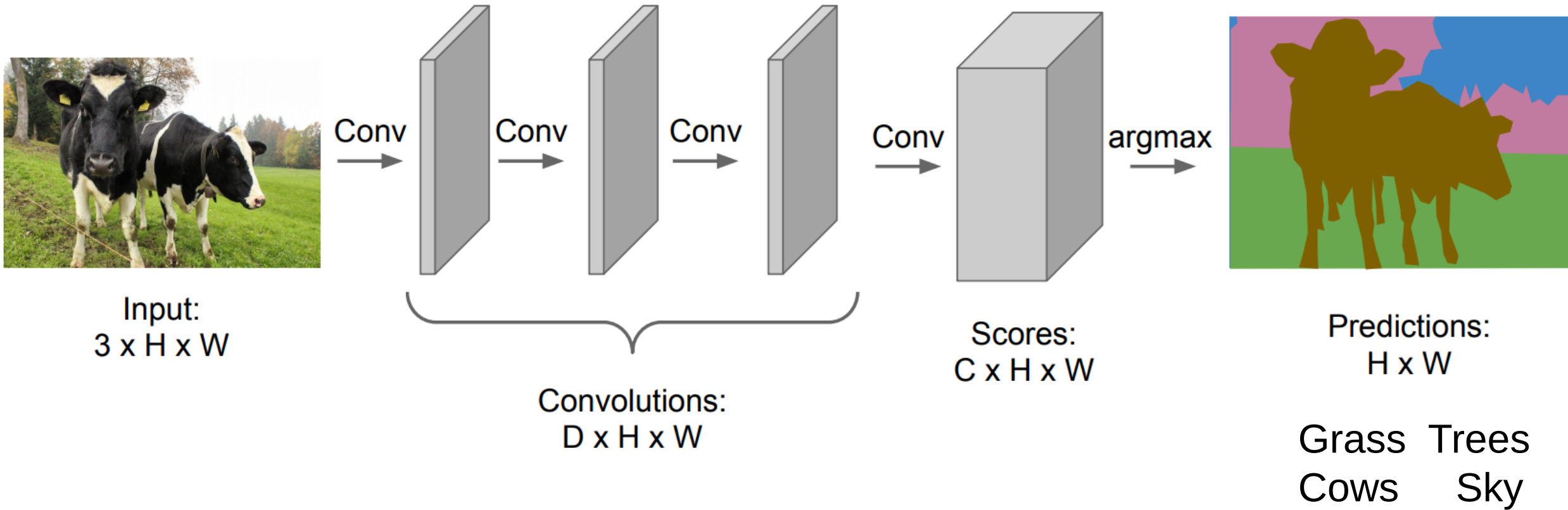


Bacteria Segmentation – Threshold at 70



Semantic Segmentation – Fully Convolutional

Design a network as a bunch of convolutional layers to make predictions for pixels all at once!

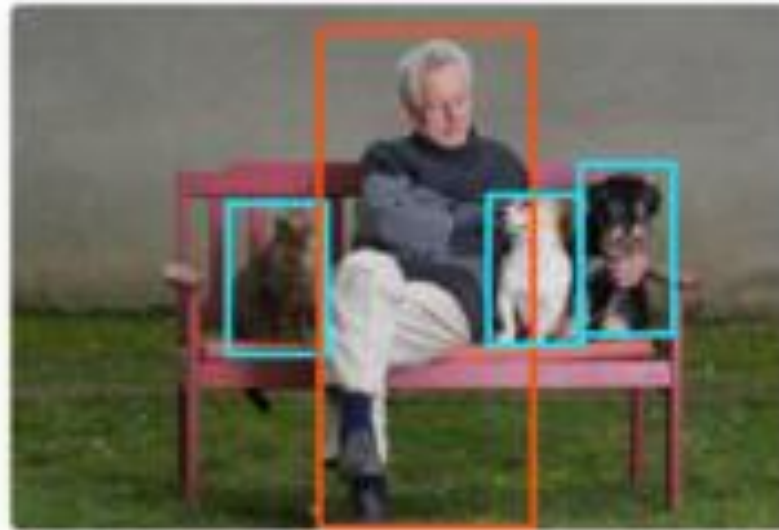


Instance Segmentation

PERSON, CAT, DOG



(A) Classification

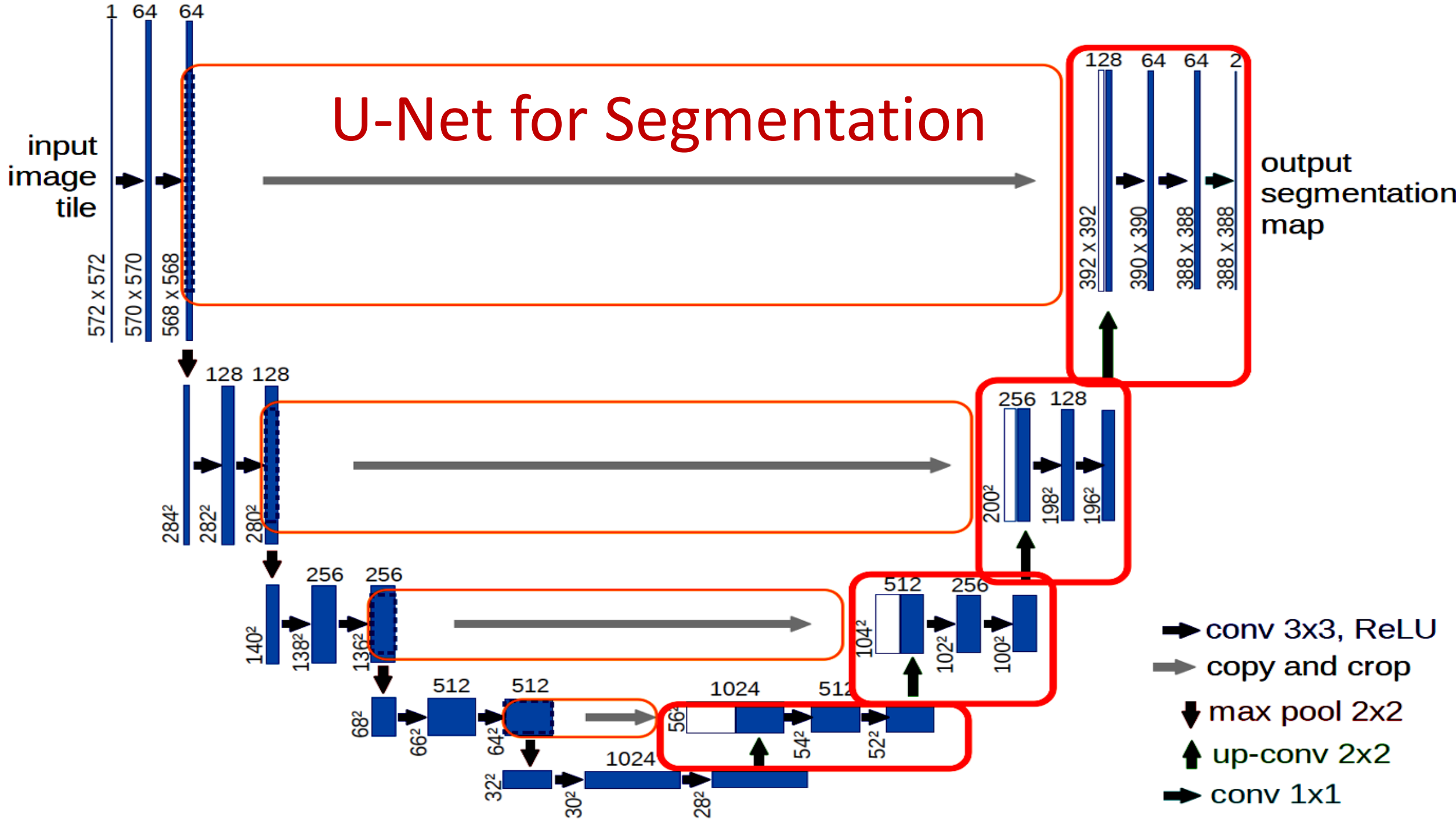


(B) Detection

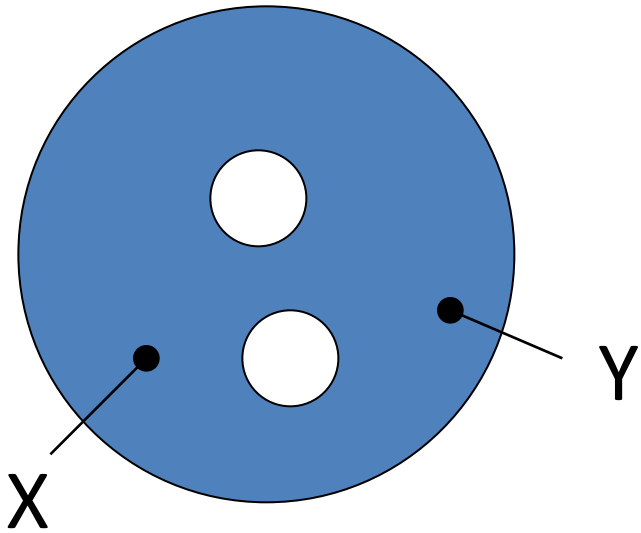


(C) Segmentation

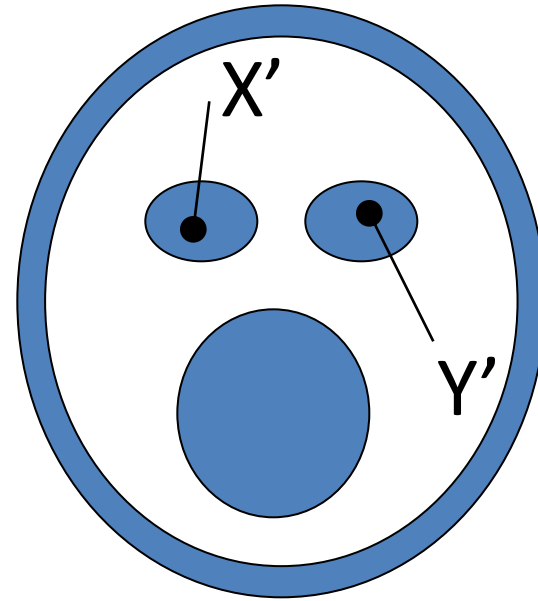
Cat Person
Dog1 Dog2



Connectivity



X and Y in same object
are connected

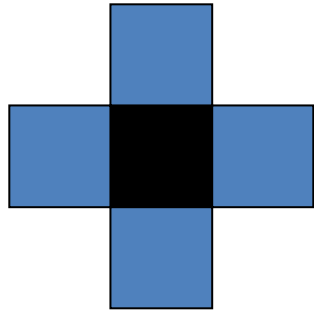


X' and Y' in same object
are NOT connected

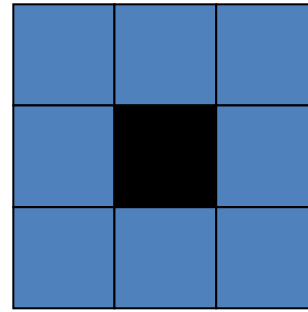
Notation: if a and b are connected, we write $a \approx b$

Pixel Neighborhoods and Connectivity

Neighbors for discrete images: 2 definitions



4-neighbors

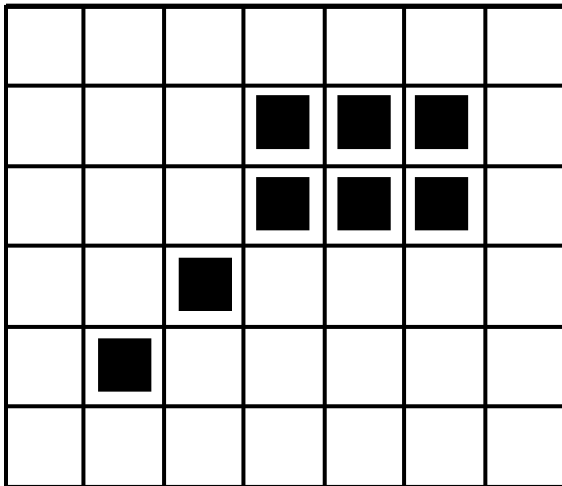


8-neighbors

- A pixel is connected to its (4 or 8) neighbors
- Reflexive: $x \approx x$ for all x
- Symmetric: if $x \approx y$, then $y \approx x$
- Transitive: if $x \approx y$ and $y \approx z$, then $x \approx z$

Connected Components

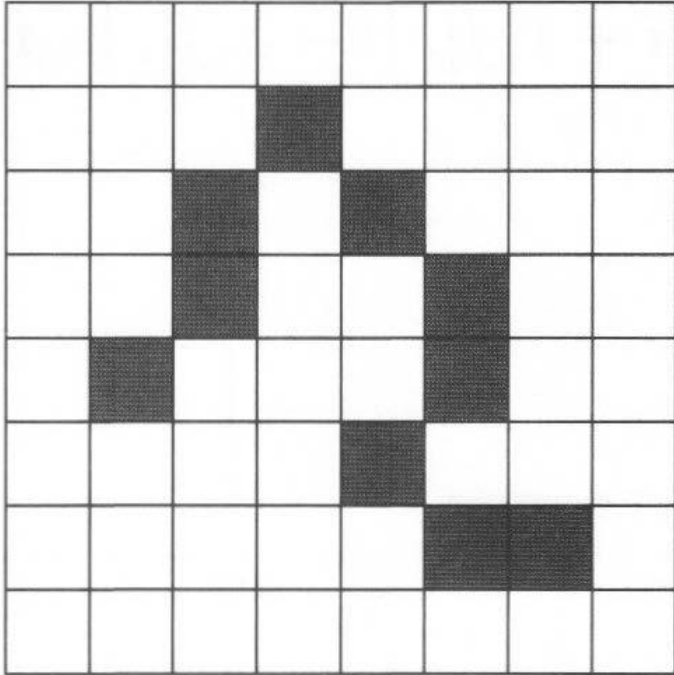
- Separate definitions for **4/8** neighborhoods.
- A set of pixels S is a Connected Component if for every two pixels (x_1, y_1) , (x_2, y_2) in S there is a path between them such that every two successive pixels in the path
(i) are in S and (ii) are **4/8** neighbors.



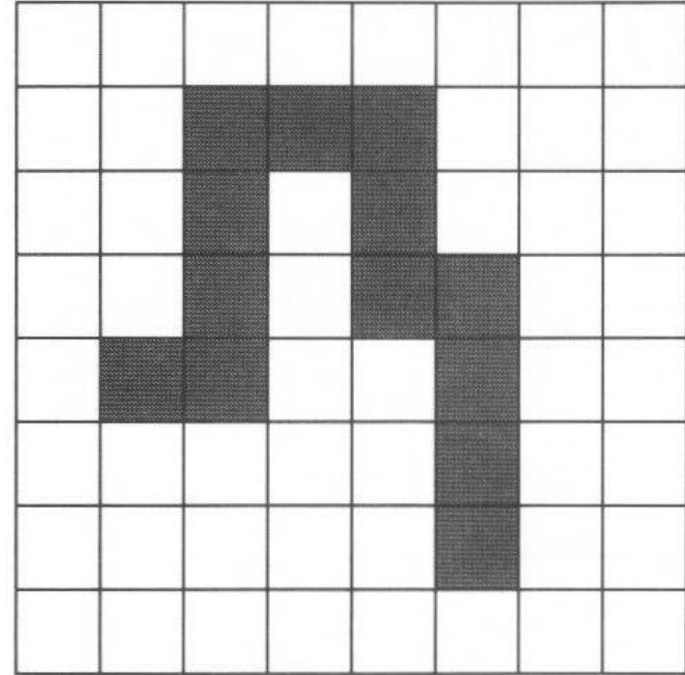
8-connectivity: 1 black component

4-connectivity: 3 black components

8/4 Path Examples

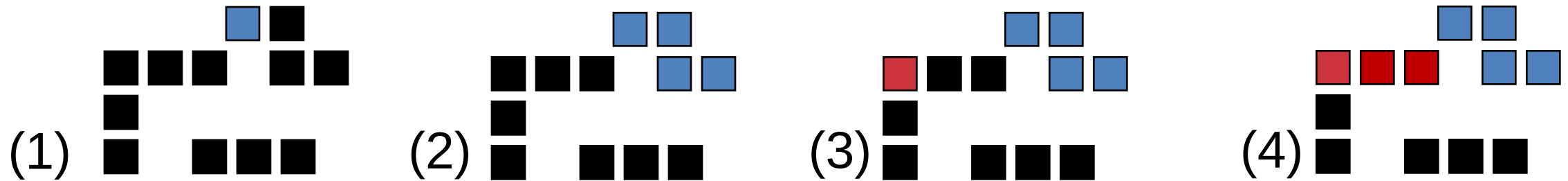


8-path



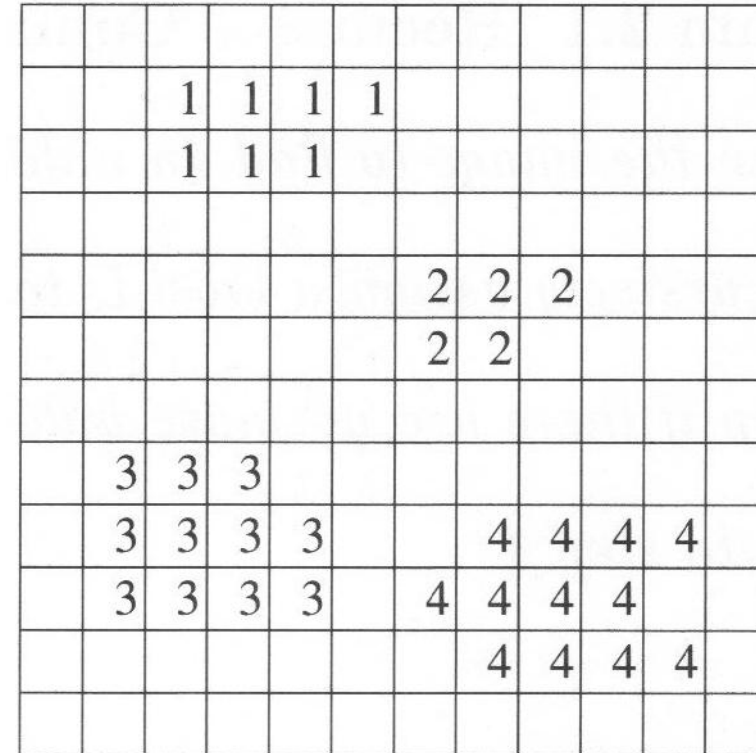
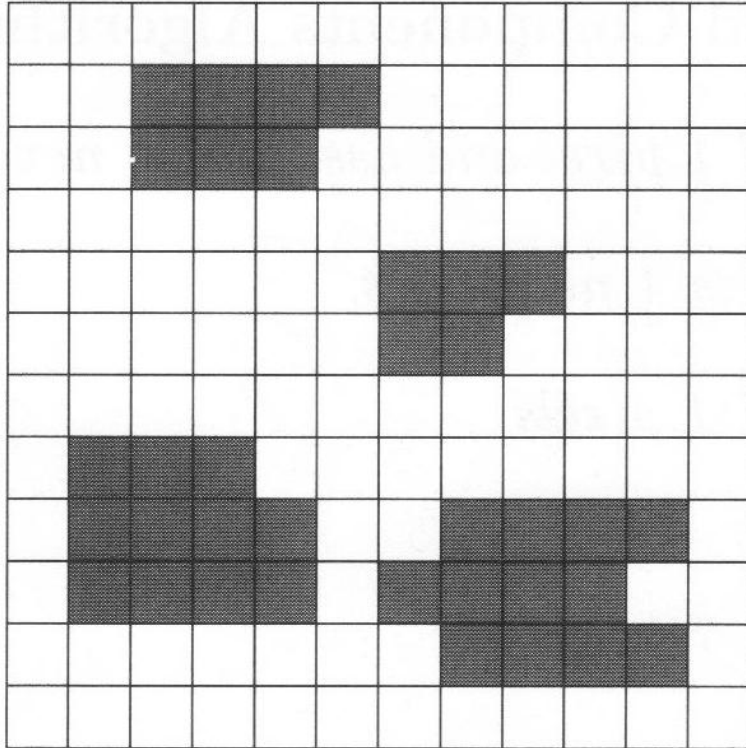
4-path

Connected Components – One Pass Scanning



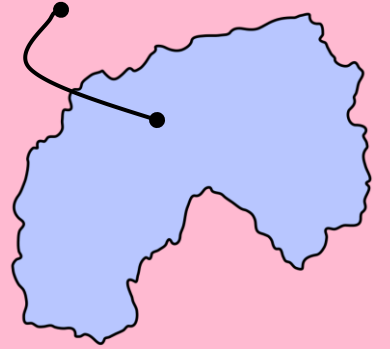
- 1) Scan image until reaching first uncolored pixel. Set color.
- 2) Continue scan, color all 4/8-neighbors of colored pixels
- 3) Start a new color on an uncolored pixel that is not a neighbor of existing color
- 4) Note two colors as equivalent when becoming neighbors
- 5) Mark all equivalent colors as one component (union-find)

Final Connected Components



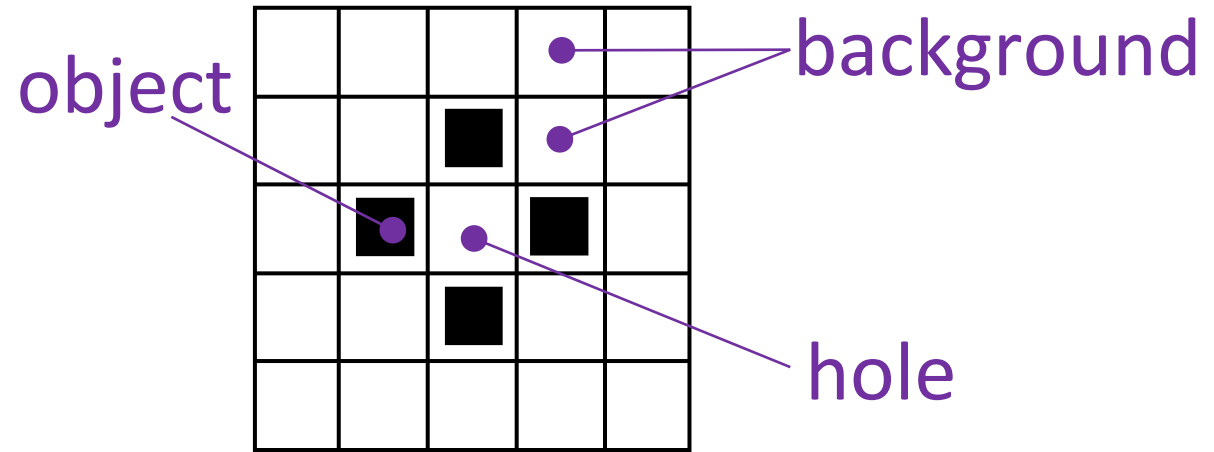
The Jordan Curve Theorem (Topology)

- Every closed curve divides the plane into an "interior" region and an "exterior" region
- Every continuous path connecting a point of one region to a point of the other intersects with that curve somewhere.



Objects and Backgrounds

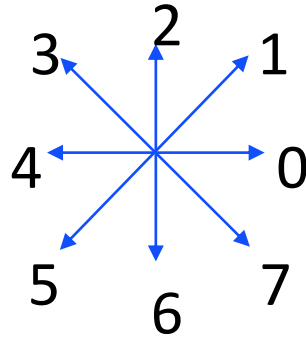
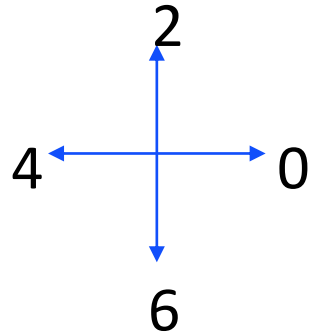
- 8-connectivity:
 - 1 black component (object)
 - 1 white component (background)
- 4-connectivity:
 - 4 black components
 - 2 white components (background & hole)
- Conflict with Jordan Curve Theorem!
 - One black component should block hole from connecting to background
 - With 4 black components, hole should connect to background
- A possible approach:
 - Use different connectivity for objects and background



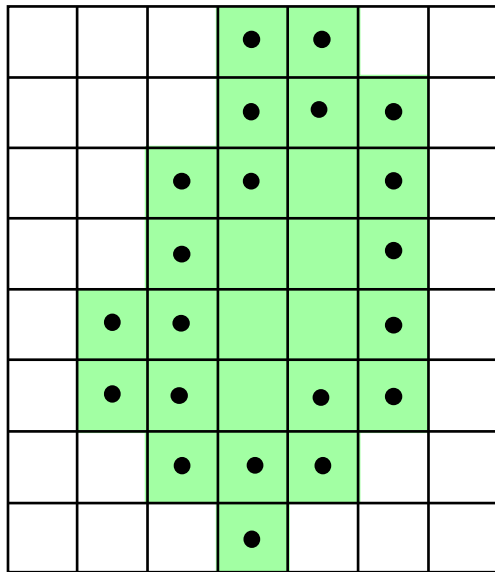
Borders

- Border = the set of all pixels in an object C that have a neighbor pixel in the background.
 - Use neighbor-8 if C is 4-connected
 - Use neighbor-4 if C is 8-connected

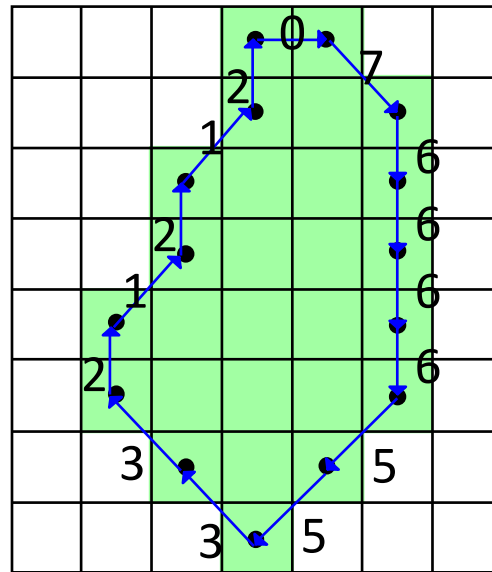
Chain Codes for Curves and Borders



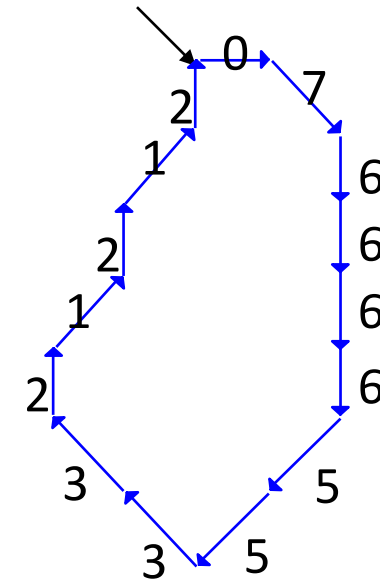
Chain code: Starting Point
+ directions to next point



8-neighbor border



4-neighbor border

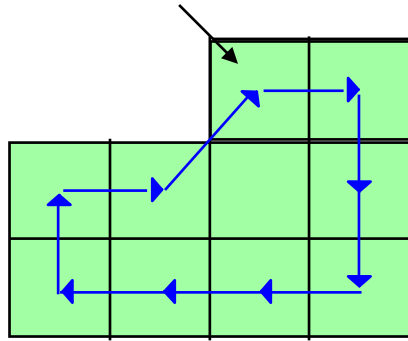


(4,1) 076666553321212

Invariance of chain code

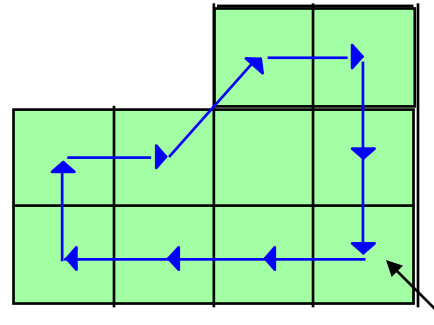
Translations, Starting point, 90° Rotations

Natural invariance to translation. Not for Rotation.



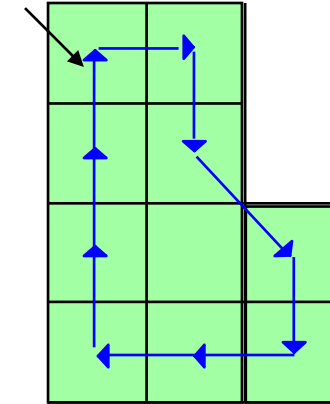
Chain-code: 066444201

Curvature: 202002271



444201066

002271202



067644222

271202002

Curvature = Differences in chain code values mod 8:

0 – Go Straight; 2-Turn right 90°; 7 – Left 45°; 1- Right 45°

Invariance to 90° rotation & starting point:

Cyclic shift giving the smallest number:

002271202

010664442

Chain Code Questions

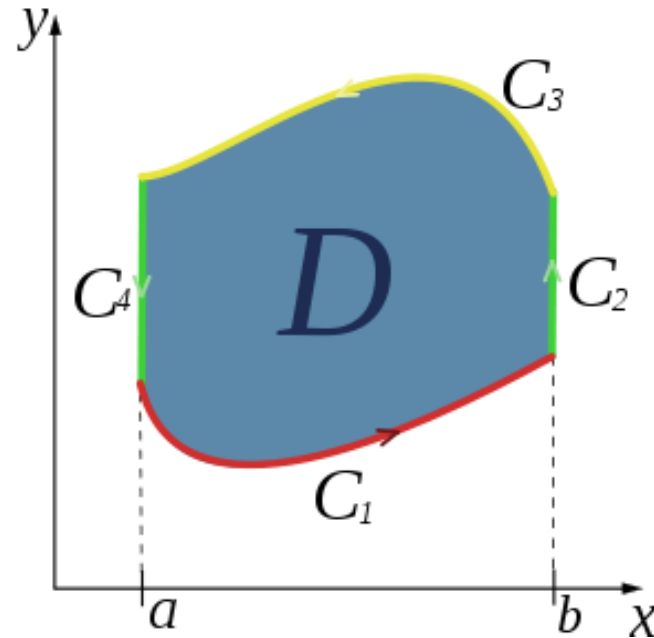
1. What are the benefits of Chain Code over raster?
2. Given a binary image, how can we generate the chain code of the boundary of the black objects?
3. How can we determine that a chain code is a closed curve, without drawing the pixels?
4. How can we determine the area of the shape enclosed by a chain code, without drawing the shape?

Sample application for Chain Code

- Makers of “LAVI” aircraft wanted real-time measurement of wing vibration during flight, by transmitting to the ground a video of the wing as seen from a video camera at the aircraft.
- The video bandwidth was too high for the radio channel.
- A proposed solution: Compute edge of wing, and transmit chain code of edge. Reduction by a factor greater than 1000
- This is the case for any information that can be encoded in lines

Green's Theorem

$$\text{Area} = \oint_C x \, dy = - \oint_C y \, dx = \frac{1}{2} \oint_C x \, dy - y \, dx$$



The area D is the area under C_3 minus the area under C_1

Distances

- Given 2 points $P=(x,y)$, $Q=(u,v)$

- Euclidean** Distance

$$d_e(P,Q) = \sqrt{(x-u)^2 + (y-v)^2}$$

- City Block (Manhattan)** Distance

$$d_4(P,Q) = |x-u| + |y-v|$$

- Chessboard (King)** Distance

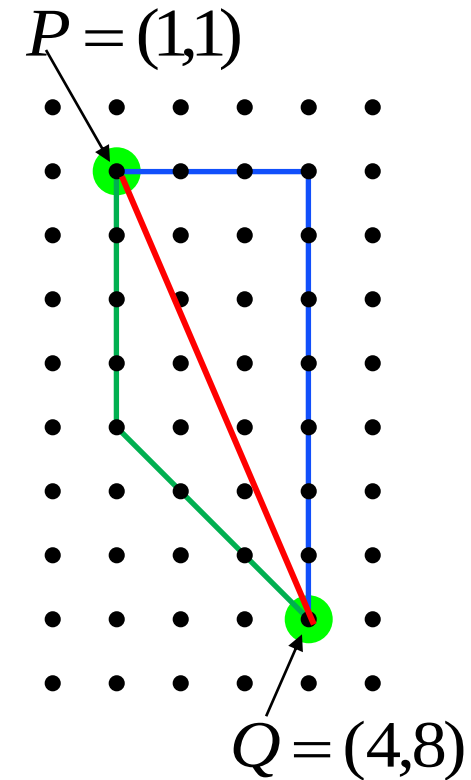
$$d_8(P,Q) = \max\{|x-u|, |y-v|\}$$

- Example: $P=(1,1)$; $Q=(4,8)$

$$d_8 = 7; \quad d_e = \sqrt{3^2 + 7^2} \approx 7.6; \quad d_4 = 10;$$

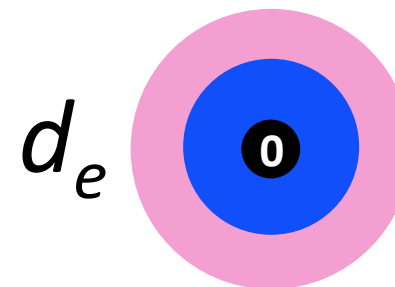
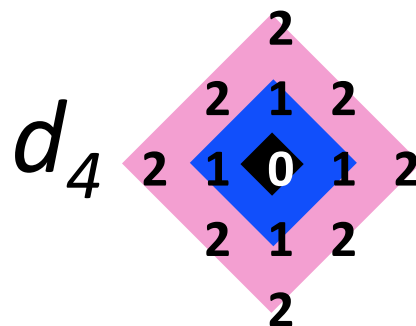
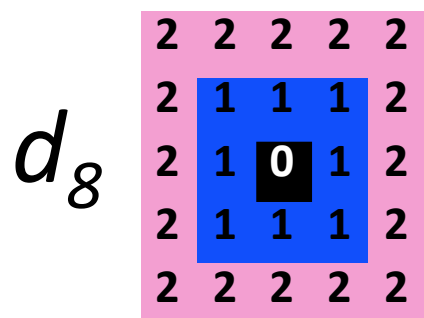
- In General: $d_8 \leq d_e \leq d_4$;

$$\sqrt{2} \cdot d_8 \geq d_e \geq d_4 / \sqrt{2}$$



Distance Properties

- Circle: all points having an equal distance



d_e , d_4 , and d_8 are metrics:

Positivity

$$d(P, Q) \geq 0, \quad \text{Only } d(P, P) = 0$$

Symmetry

$$d(P, Q) = d(Q, P)$$

Triangle Inequality

$$d(P, Q) \leq d(P, R) + d(R, Q)$$

Different Distance Measures

			3			
		3	2	3		
	3	2	1	2	3	
3	2	1	0	1	2	3
	3	2	1	2	3	
		3	2	3		
			3			

City Block d_4

3	3	3	3	3	3	3
3	2	2	2	2	2	3
3	2	1	1	1	2	3
3	2	1	0	1	2	3
3	2	1	1	1	2	3
3	2	2	2	2	2	3
3	3	3	3	3	3	3

Chessboard d_8

			3			
	$\sqrt{8}$	$\sqrt{5}$	2	$\sqrt{5}$	$\sqrt{8}$	
	$\sqrt{5}$	$\sqrt{2}$	1	$\sqrt{2}$	$\sqrt{5}$	
3	2	1	0	1	2	3
	$\sqrt{5}$	$\sqrt{2}$	1	$\sqrt{2}$	$\sqrt{5}$	
	$\sqrt{8}$	$\sqrt{5}$	2	$\sqrt{5}$	$\sqrt{8}$	
			3			

Euclidean d_e

Distance Questions

1. Prove that d_e , d_4 , and d_8 are metrics
2. Prove: $d_8 \leq d_e \leq d_4$; $\sqrt{2} \cdot d_8 \geq d_e \geq d_4 / \sqrt{2}$
3. Prove that $[d_e]$ (rounded d_e) is not a metric

Distance Maps / Distance Transforms

Value at each pixel is the distance from a given set of points (distance to the closest point in the set).

Example: 3 points below. Greylevel = Distance, Black=0.

Blue=25

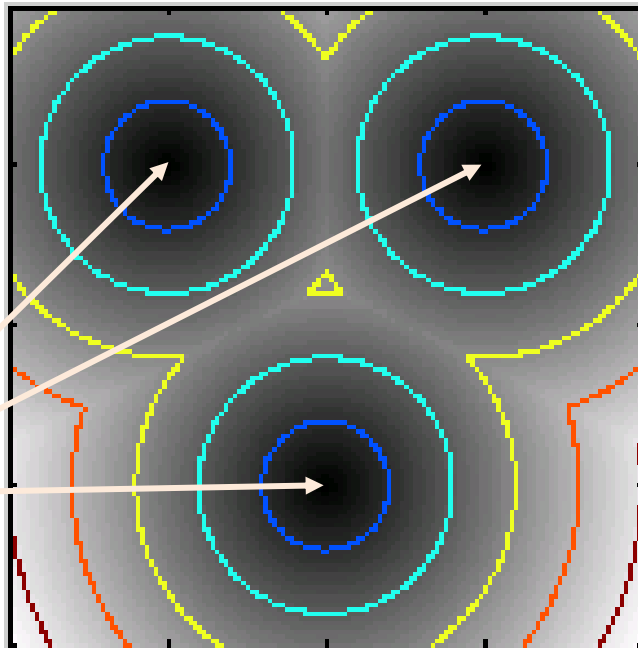
Cyan=50

Yellow=75

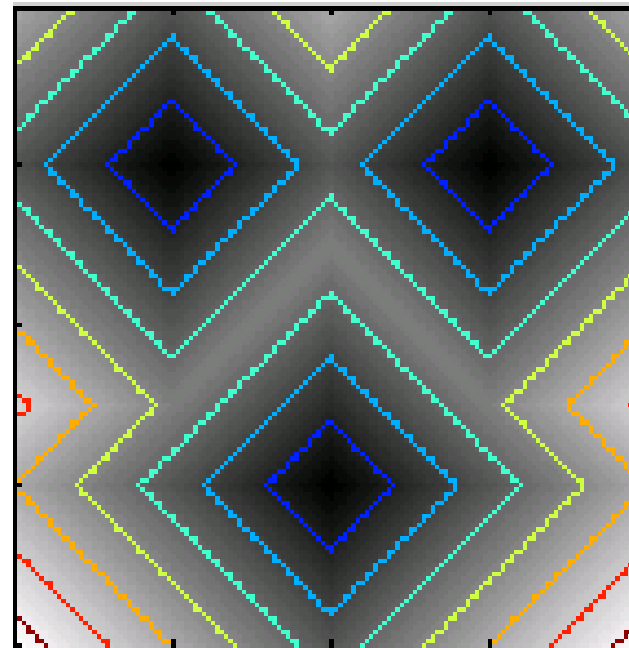
Red=100

Brown=125

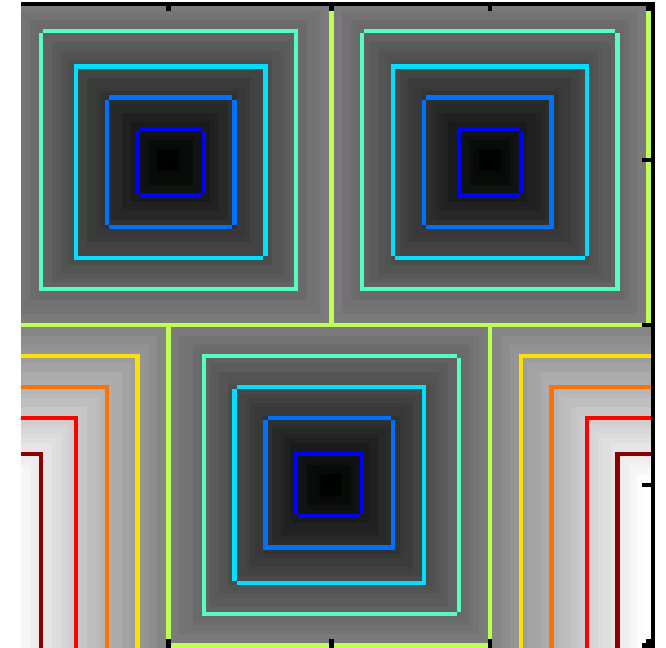
Euclidean



City Block



Chessboard



Distance Maps / Distance Transforms

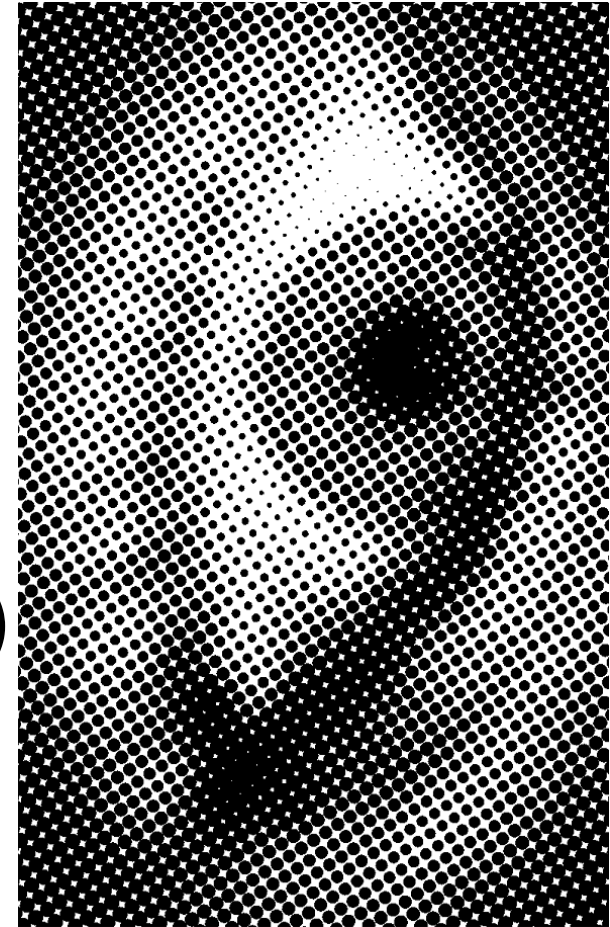
- Many fast algorithms exist to compute distance transform, Euclidean distance is more challenging.
- Another Distance Transform example:
- Distance to closest boundary point
- Greylevel in image below = Distance

What distance was used?



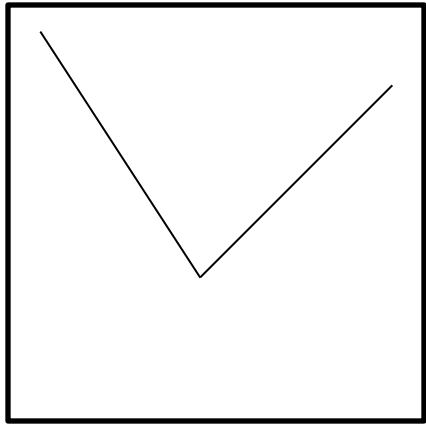
Distance Transform for Oilfields (Schlumberger)

- What is the meaning of the distance transform for the image on right (1-black, 0-white)?
- Width of spaces between black points
- Real-World Application:
 - Take a piece of rock from an oil field
 - Put rock on a 3D CT Scanner (Same CT as in Hospitals)
 - Compute distance transform (1-Rock, 0-air)
 - From histogram of distance transform infer size of holes and the flow of oil in rock

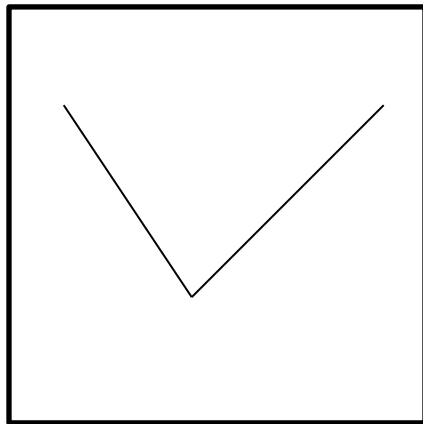


Example for Using Distance Transform

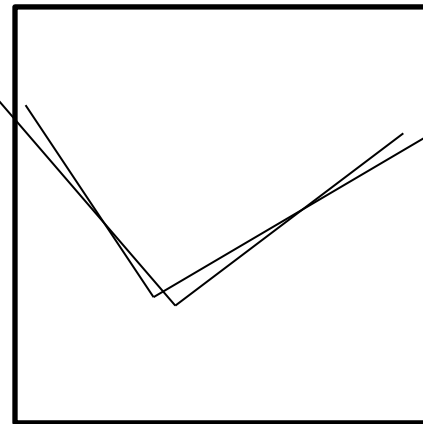
- Path planning and navigation: find high clearance paths
- Align pictures of black lines on a white background.
 - Hard to find best correlation / derivatives
 - Compute distance transform before alignment
 - Align the distance transform images



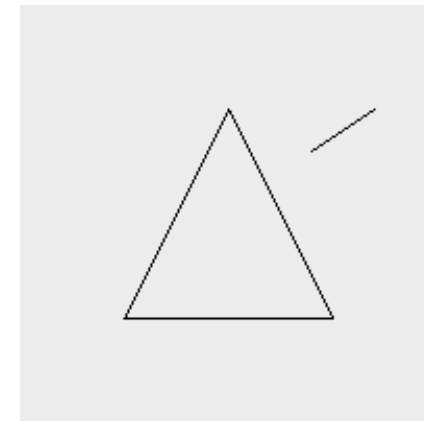
A



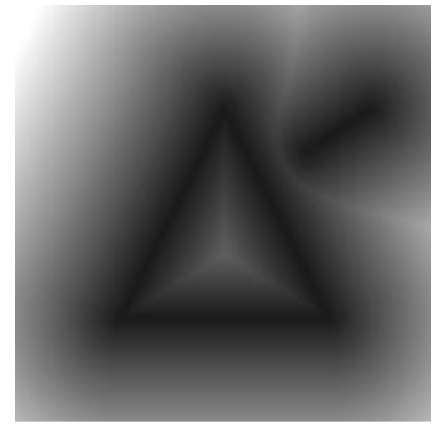
B



Best Alignment



Binary Curve



Distance Transform

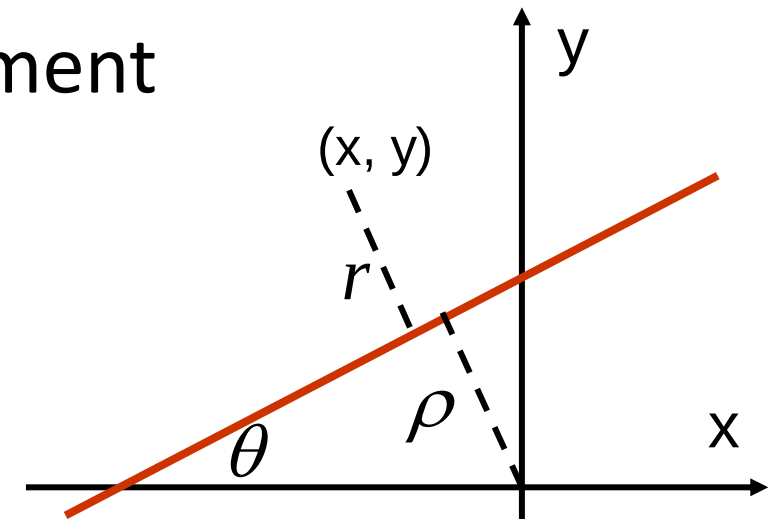
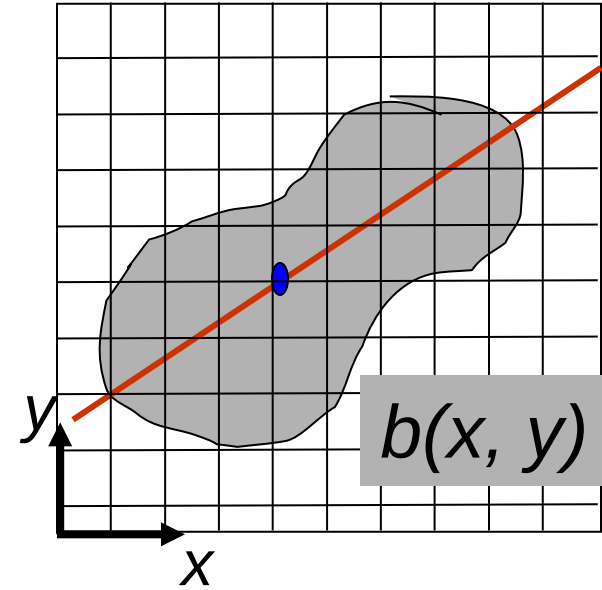
Properties of Binary Objects: Moments

- Moments for any given m, n : $\int x^n y^m b(x, y) dx dy$
- Area: Zero'th moment ($m=n=0$): $A = \sum \sum b(x, y)$
- Location of Centroid: 1st moments $(m, n) = (1, 0), (0, 1)$

– $\bar{x} = \frac{1}{A} \sum \sum x b(x, y); \quad \bar{y} = \frac{1}{A} \sum \sum y b(x, y)$

- Orientation? “major axis” that minimizes Moment of Inertia

– $E = \iint r^2 b(x, y) dx dy$



Major Axis

- 2nd moments $(m,n) = (2,0), (1,1), (1,1)$

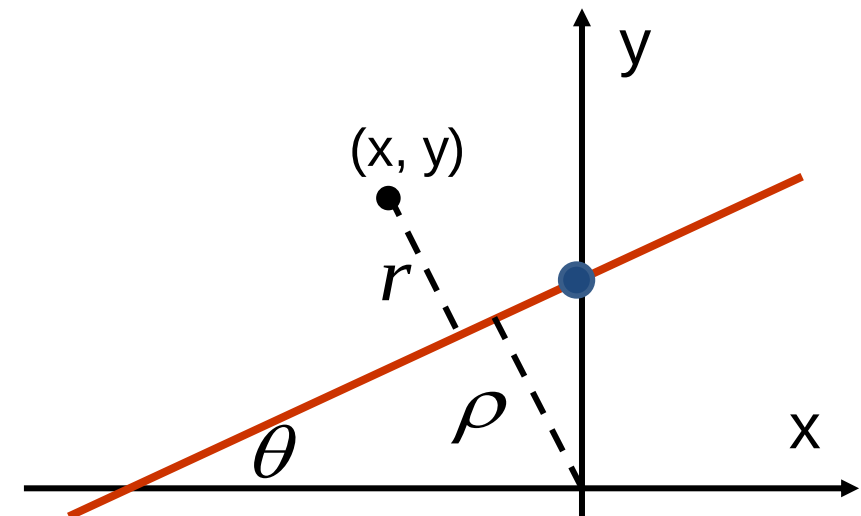
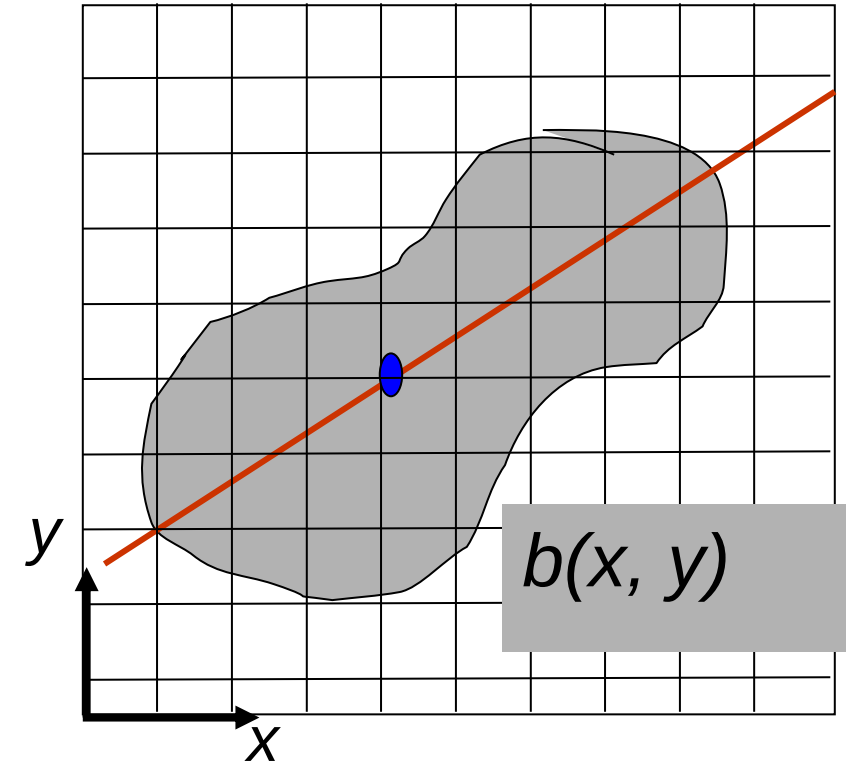
$$a' = \sum \sum x^2 b(x, y);$$

$$b' = 2 \sum \sum xy b(x, y);$$

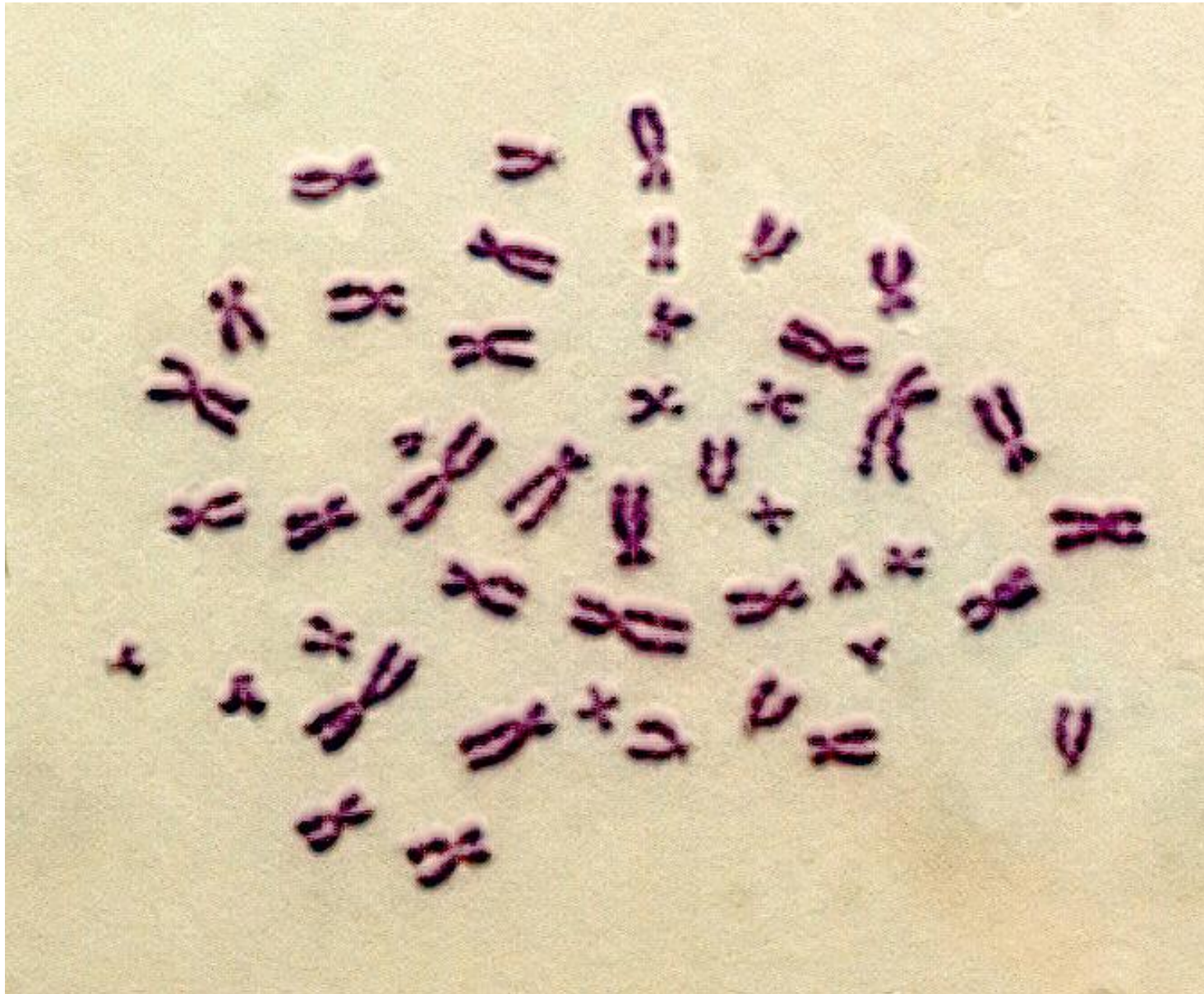
$$c' = \sum \sum y^2 b(x, y)$$

- Major axis passes through centroid
- Direction θ satisfies

$$\tan 2\theta = \frac{b'}{a' - c'}$$

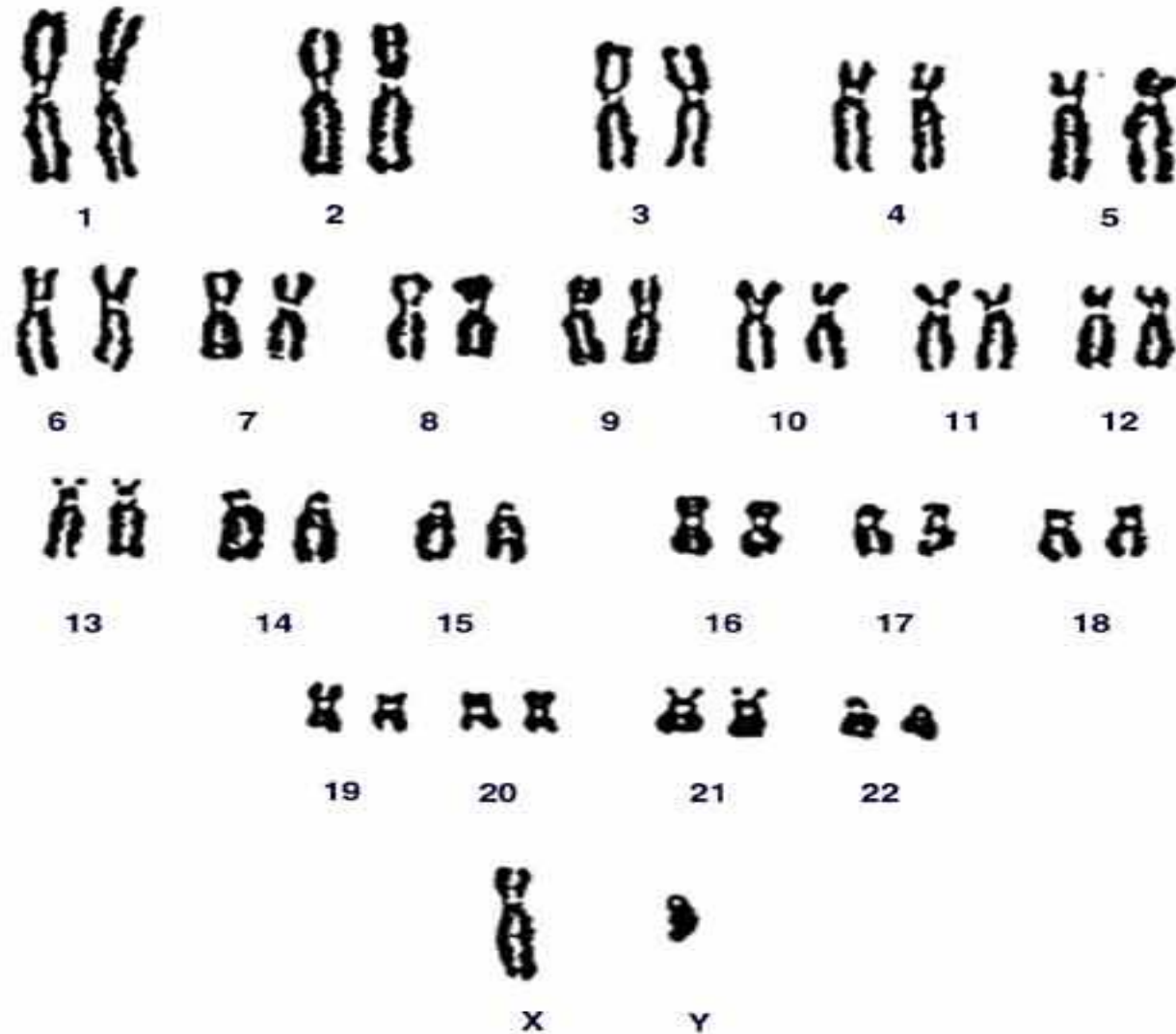


Normalizing Direction Example: Chromosomes



Organized by Size, Normalized Orientation

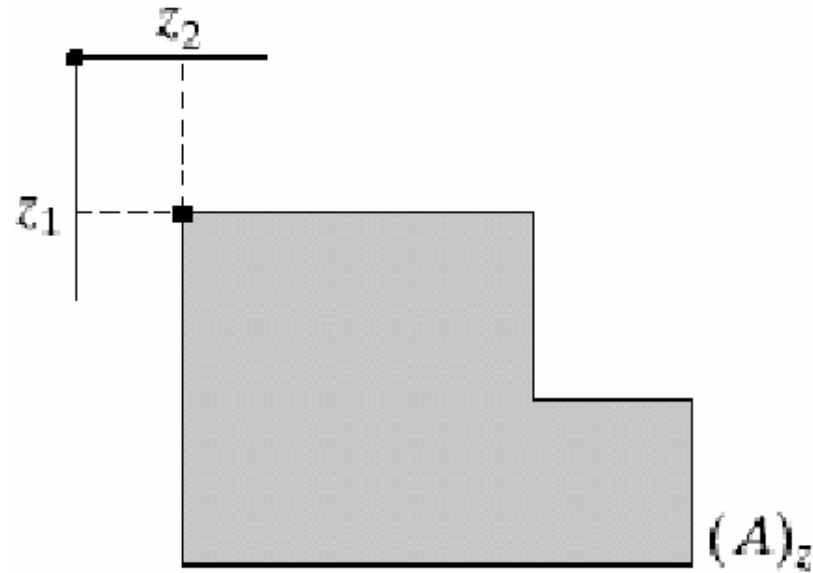
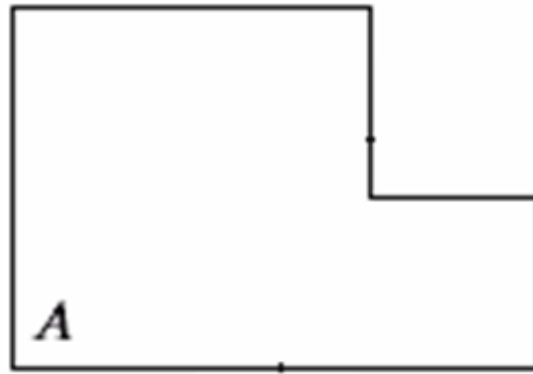
Compute Connected Components, Area, Direction



Mathematical Morphology - Translation

$$(A)_z = \{w \mid w = a + z, a \in A\} \quad z = (z_1, z_2)$$

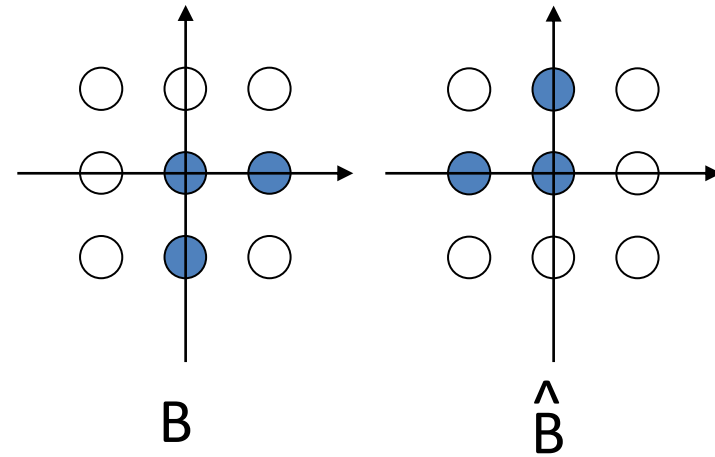
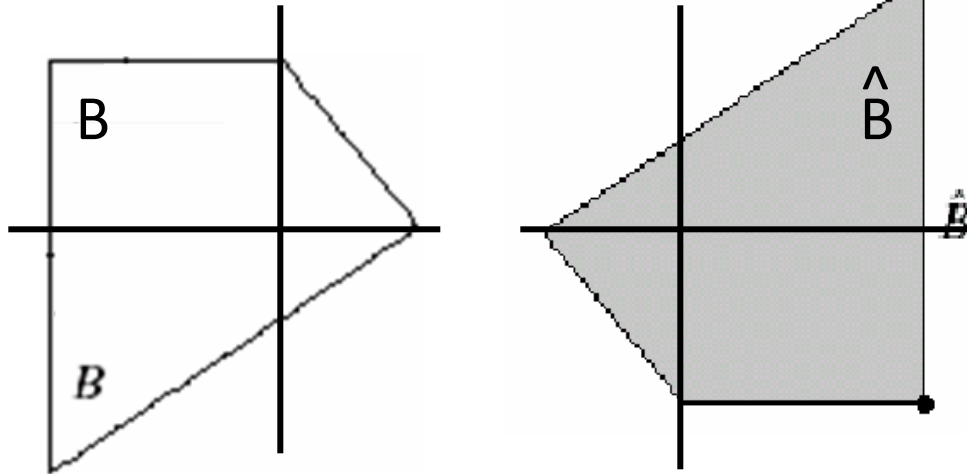
Example



Reflection

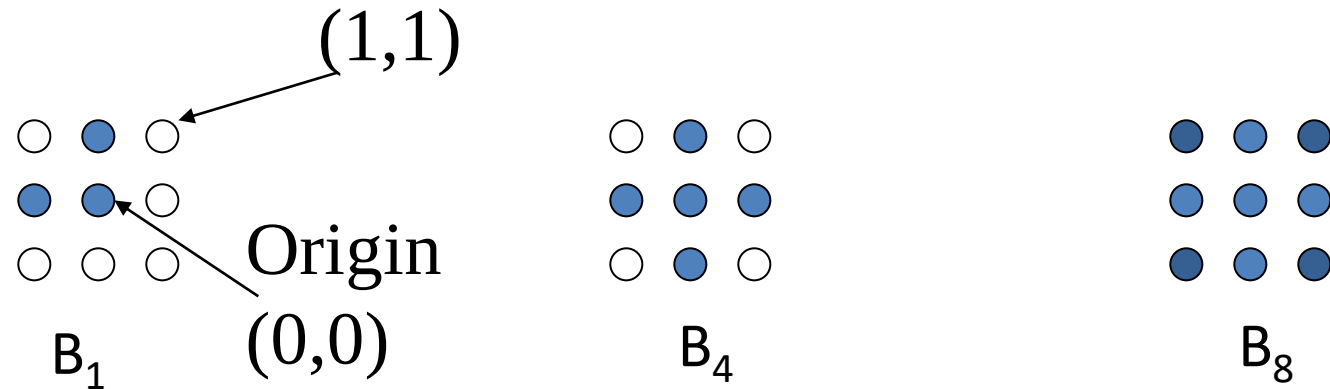
$$\hat{B} = \{w \mid w = -b, b \in B\}$$

Example



Structuring Element

- A set of local neighborhood with a specified origin



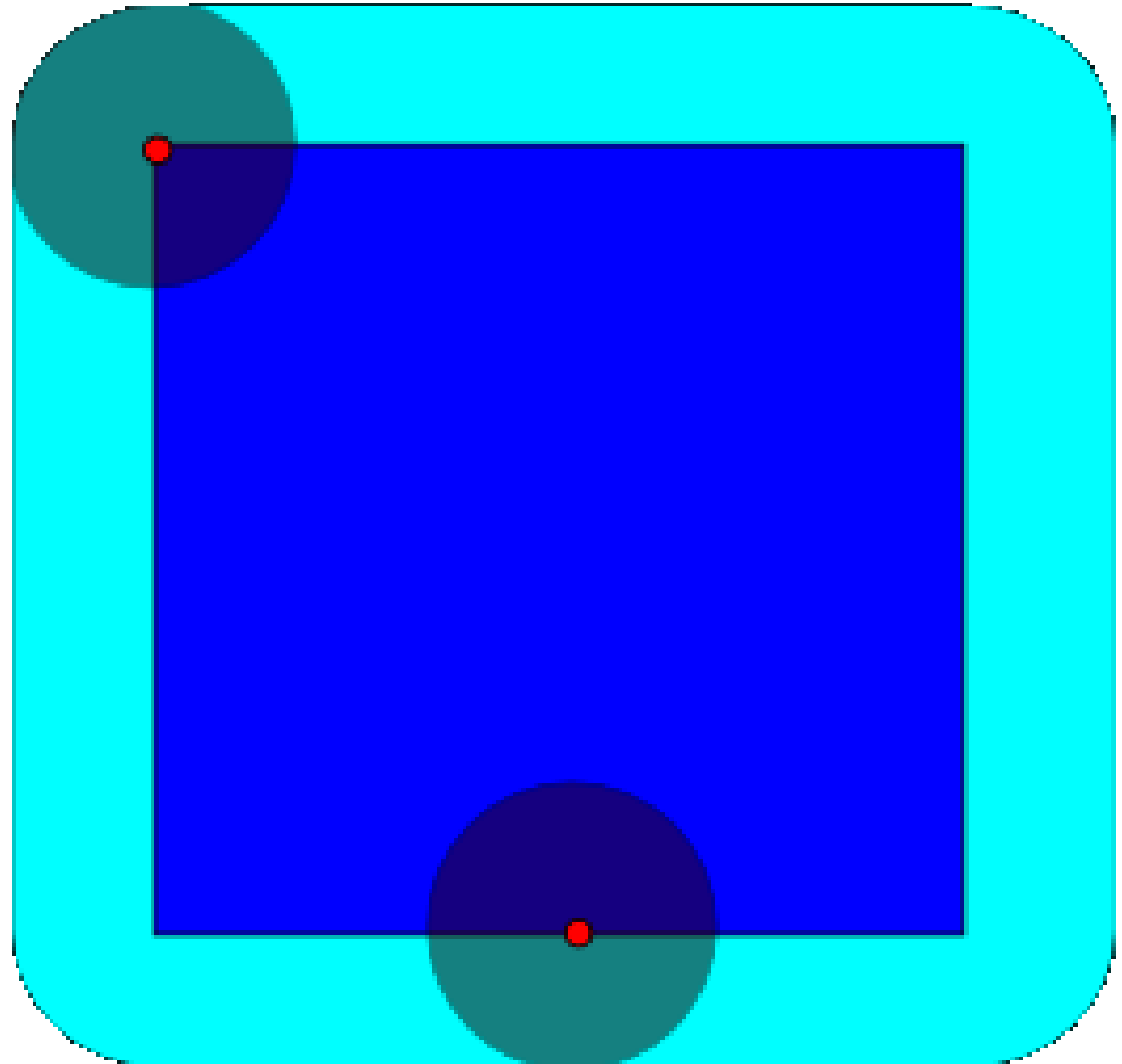
$$B_1 = \{(0,0), (-1,0), (0,1)\}$$

$$B_4 = \{(0,0), (-1,0), (0,1), (1,0), (0,-1)\}$$

$$B_8 = \{(0,0), (-1,0), (0,1), (1,0), (0,-1), (-1,1), (1,1), (1,-1), (-1,-1)\}$$

Dilation

- Before: 
- After: 

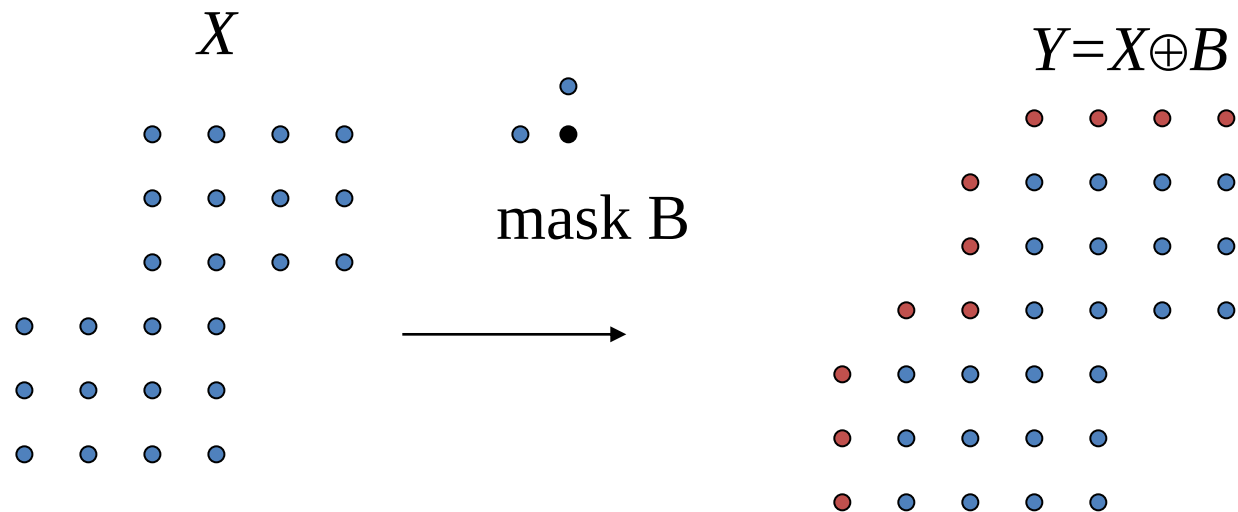


הרחבה Dilation

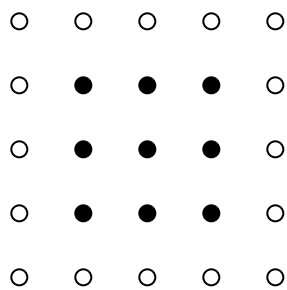
- Definition

$$Y = X \oplus B = \bigcup_{b \in B} X_b = \bigcup_{x \in X} B_x = B \oplus X$$

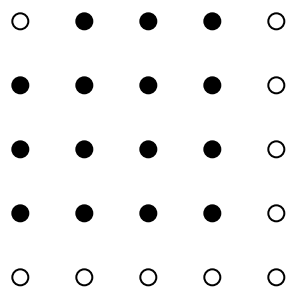
- Example



More Dilation



X



$X \oplus B$

origin

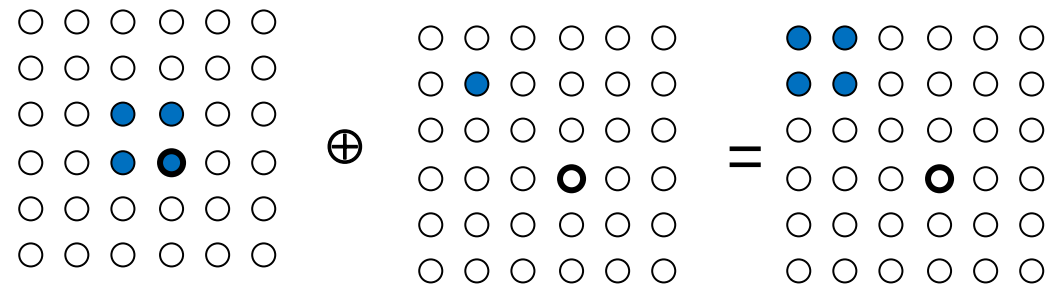
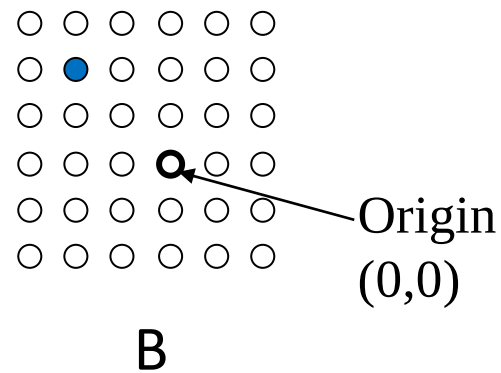


B

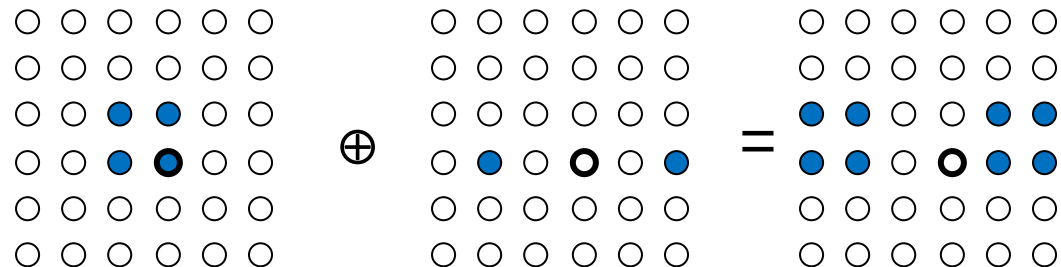
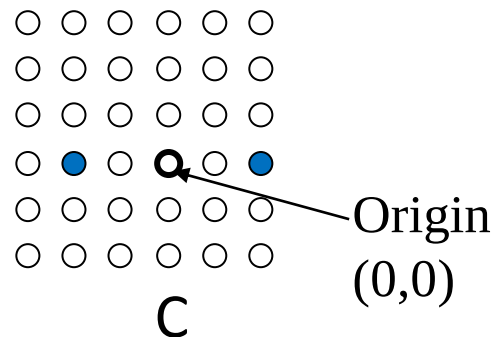
$$B = \{(0,0), (0,1), (-1,0)\}$$

More Examples: Structuring Elements

- Shift by $S \oplus B$

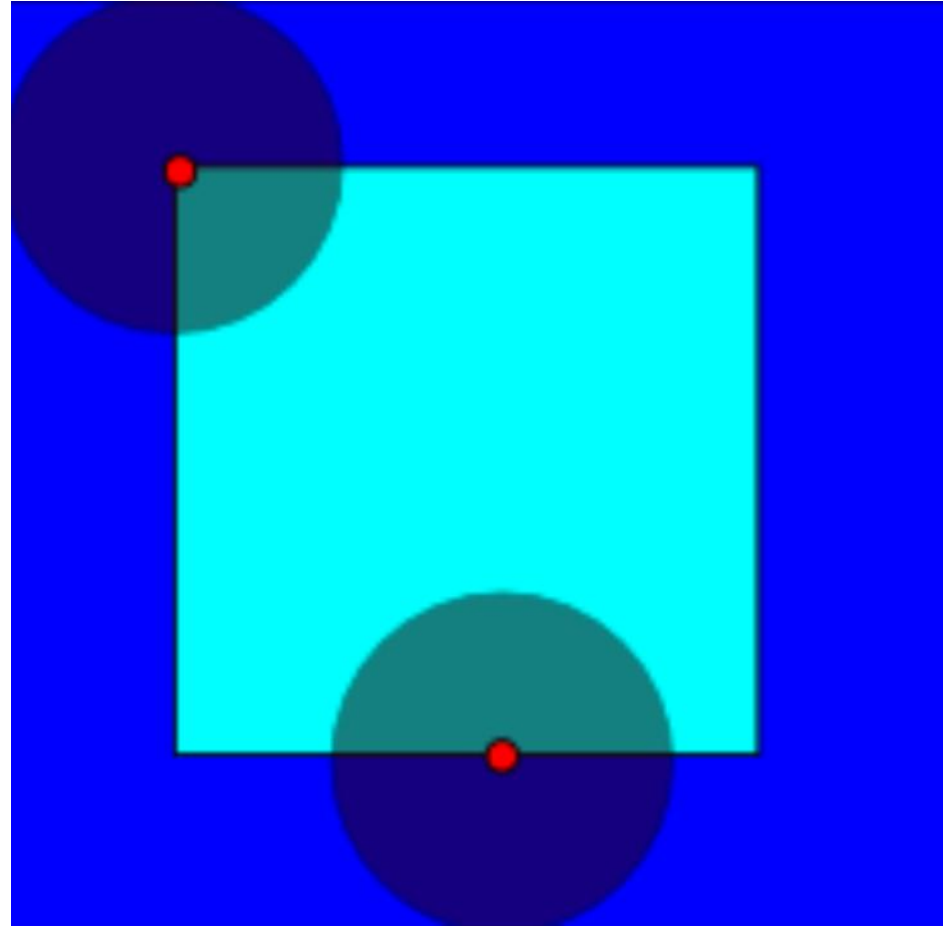


- Duplicate by $S \oplus C$



Erosion

- Before: 
- After: 

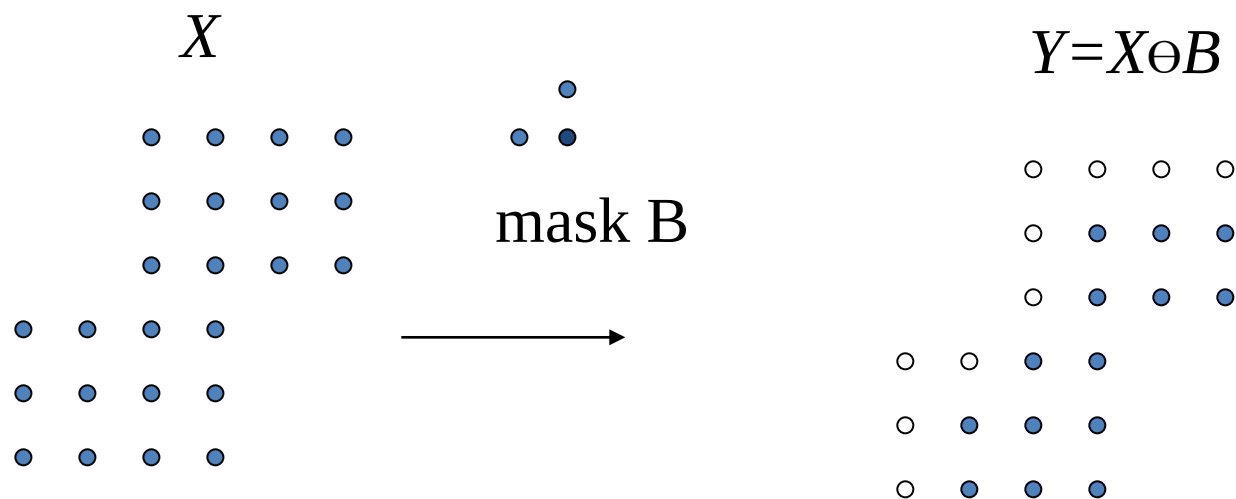


כיוון Erosion

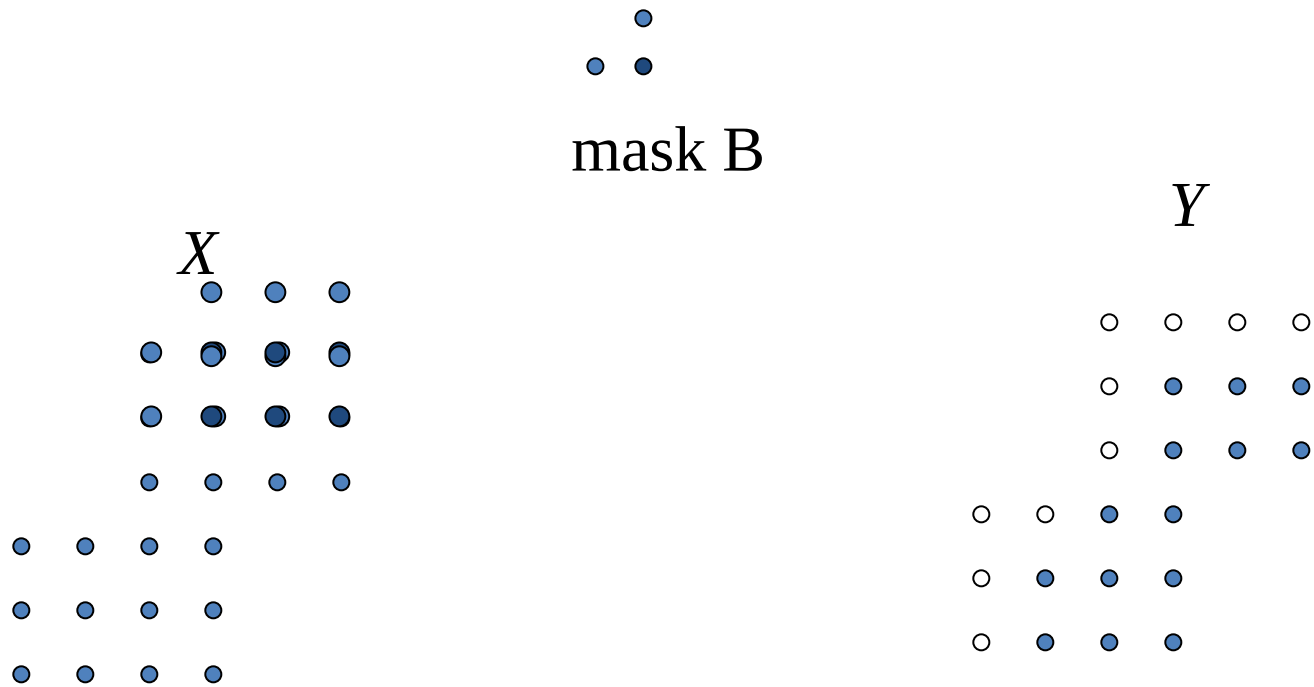
Definition

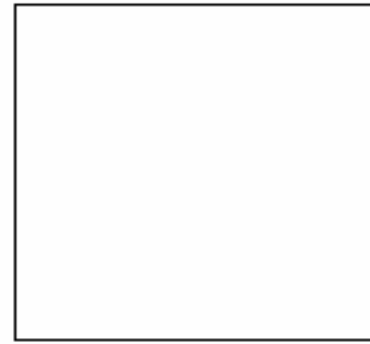
$$Y = X \ominus B = \{x : B_x \subset X\} = \{x : B_x \cap X^c = \emptyset\}$$

Example

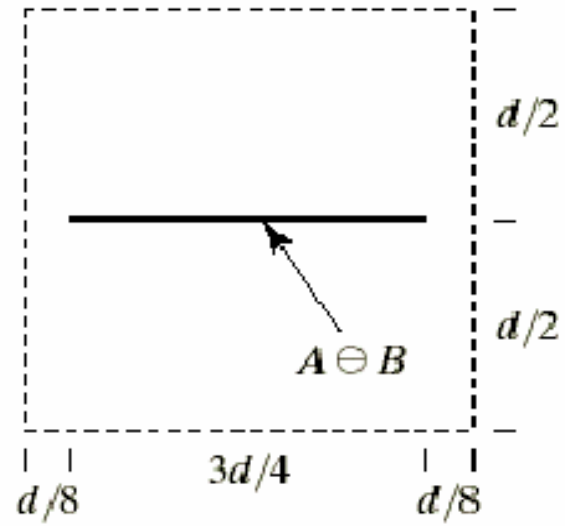
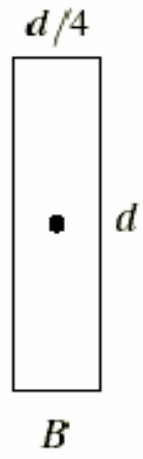
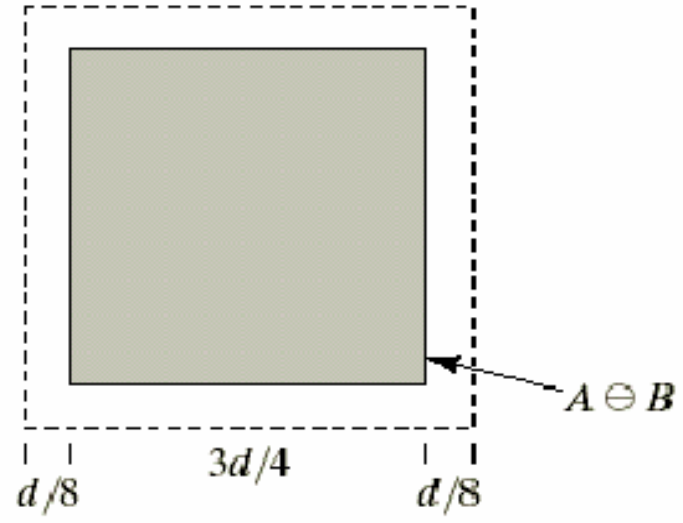
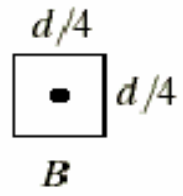


More Erosion

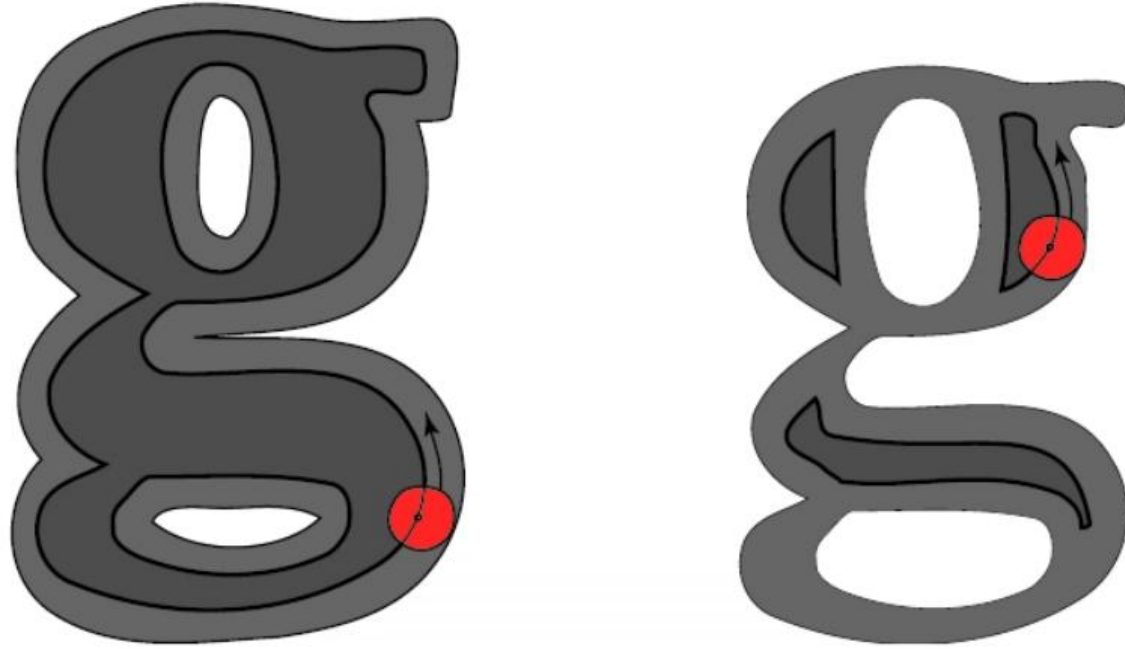




A



Example



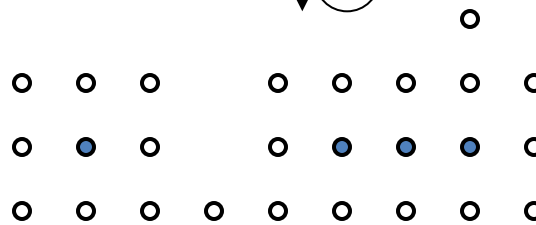
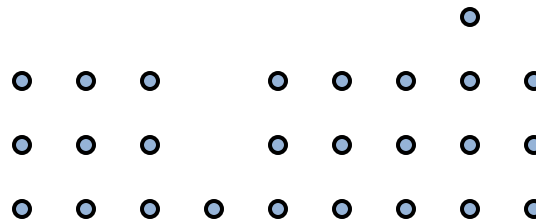
Opening: Erosion + Dilation

Definition

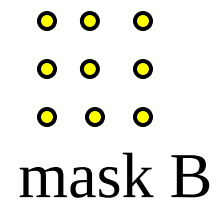
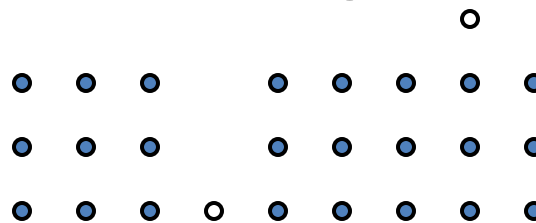
$$X \circ B = (X \ominus B) \oplus B$$

Example

X

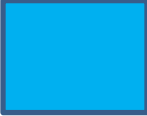


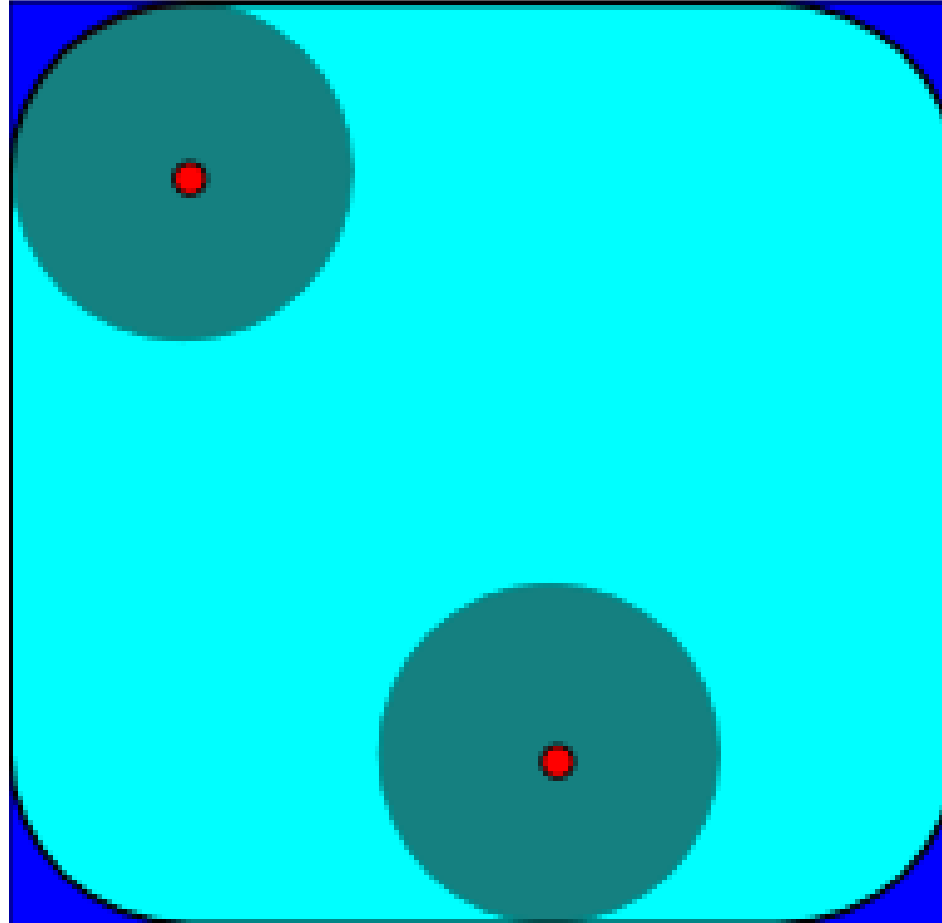
$X \circ B$



Opening

- Before: 

- After: 



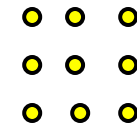
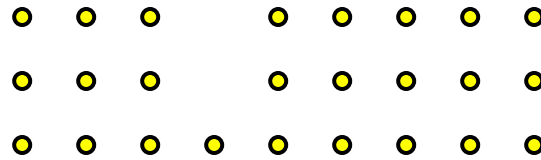
Closing: Dilation and Erosion

Definition

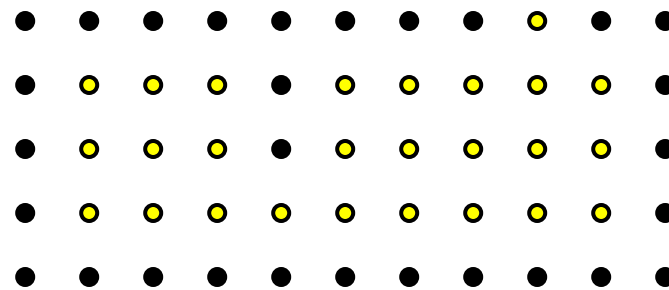
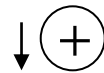
$$X \bullet B = (X \oplus B) \ominus B$$

Example

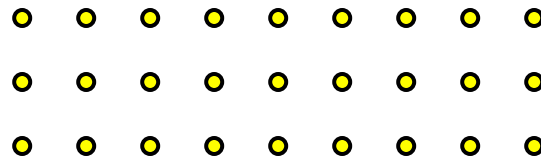
X



mask B

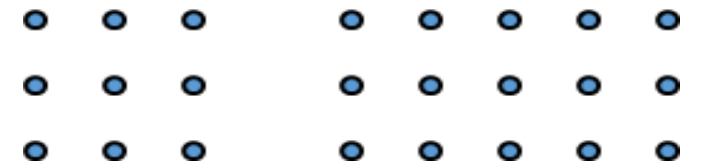


$X \bullet B$



$X \circ B$

Opening



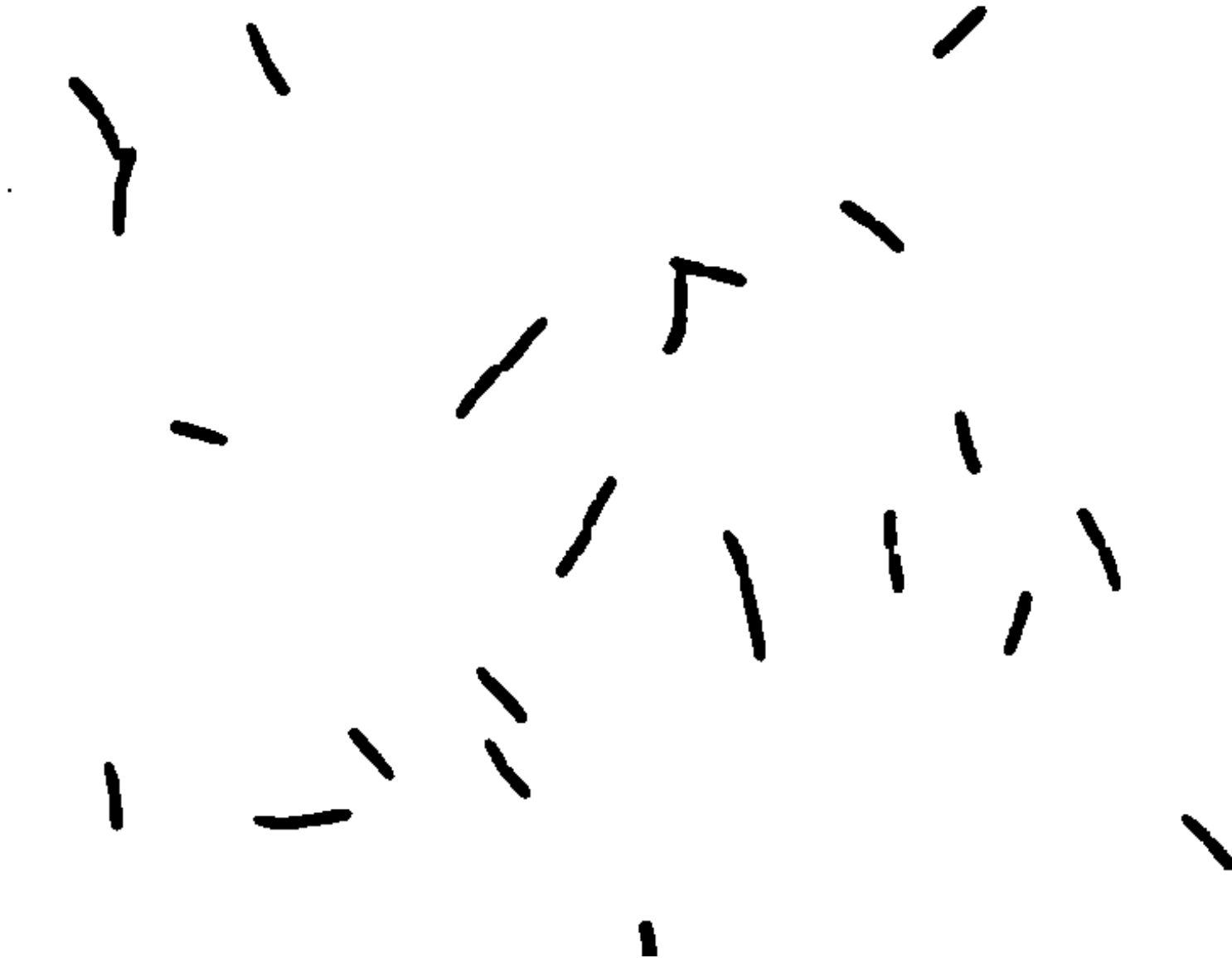
Bacteria Original



Dilation



Closing = Dilation + Erosion



Bacteria Original



Properties of Opening and Closing

Opening

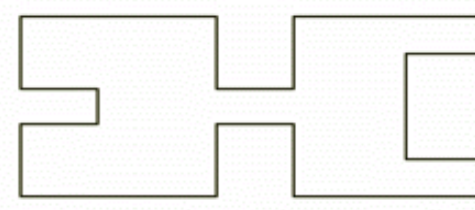
- $X \circ B \subseteq X$
- $X \subseteq Y \Rightarrow X \circ B \subseteq Y \circ B$
- $(X \circ B) \circ B = X \circ B$

Closing

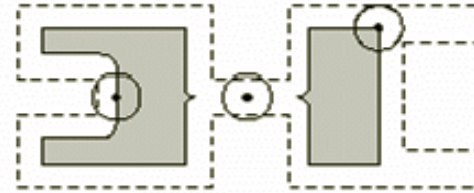
- $X \subseteq X \bullet B$
- $X \subseteq Y \Rightarrow X \bullet B \subseteq Y \bullet B$
- $(X \bullet B) \bullet B = X \bullet B$

$$(X \bullet B)^c = X^c \circ \hat{B} \qquad (X \circ B)^c = X^c \bullet \hat{B}$$

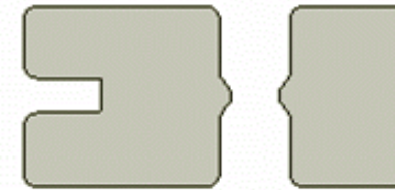
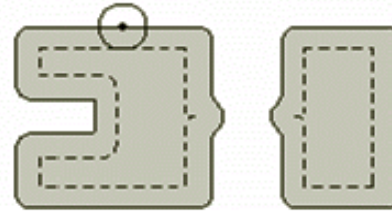
Effects of Opening and Closing



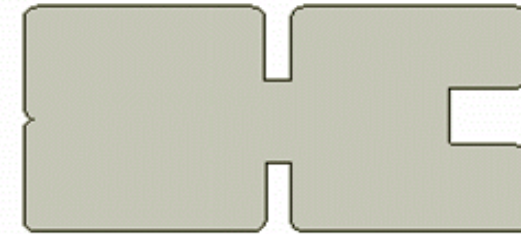
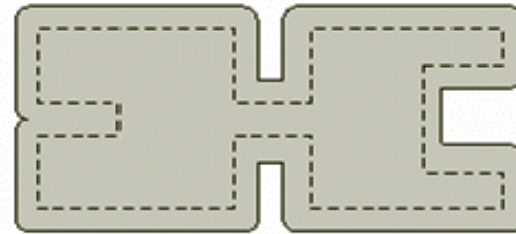
A



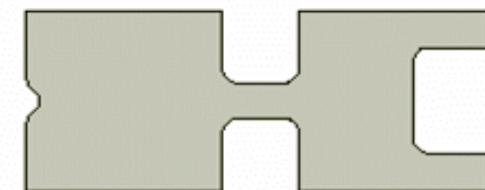
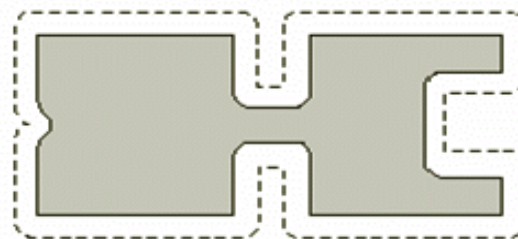
$A \ominus B$



$$A \cdot B = (A \ominus B) \oplus B$$



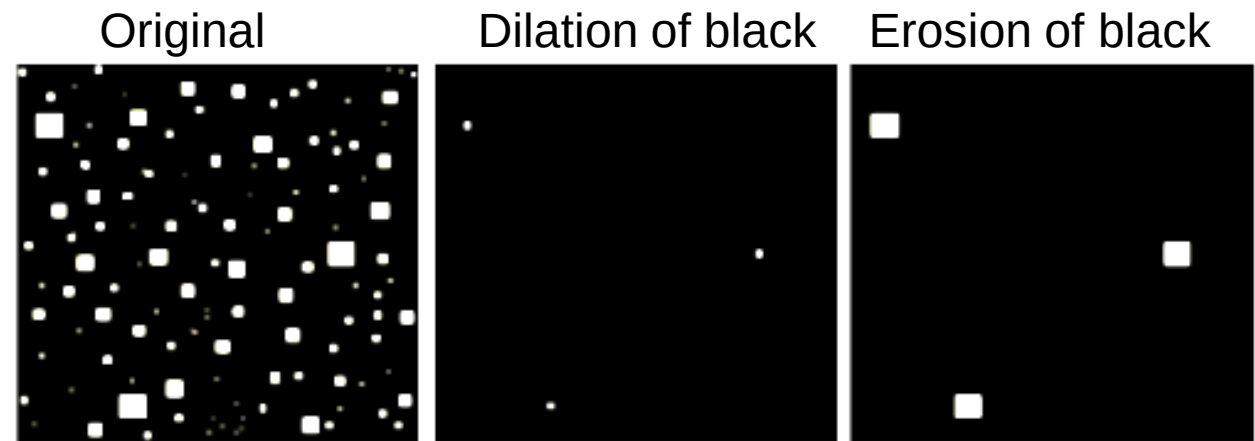
$A \oplus B$



$$A \cdot B = (A \oplus B) \ominus B$$

Simple Operations

- Removing small objects by opening
- Filling small holes by closing



- Edge Detection:

$$\partial(A) = A - (A \ominus B); \quad \text{What is } (A \oplus B) - A?$$

Grayscale Morphology: Dilation

Binary: $Y = X \oplus B = \bigcup_{b \in B} X_b = \max_{b \in B} \{X_b\}$

Grey: $Y(x, y) = \max_{b \in B} \{X(x + b_x, y + b_y)\}$



Grayscale Morphology: Erosion

$$Y = X \ominus B = \min_{b \in B} \{X_b\}$$

$$Y(x, y) = \min_{b \in B} \{X(x + b_x, y + b_y)\}$$

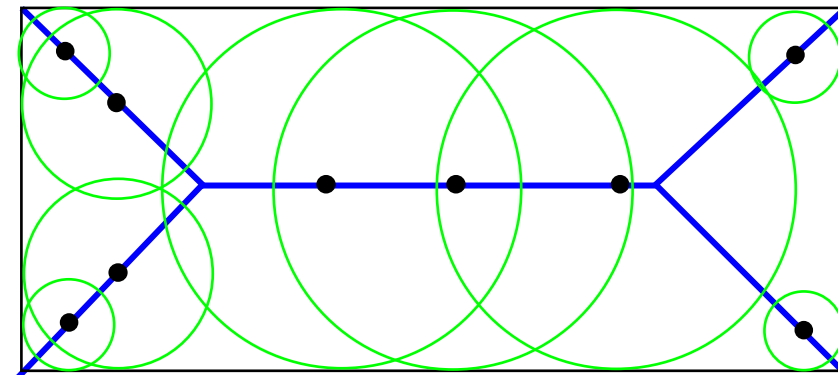


Skeleton-Medial Axis

- A skeleton point is a local maximum (4/8) in the distance transform to boundary
- Using 8-neighborhood
- Using 4-neighborhood

1	1	1	1	1	1	1	1	1	1
1	2	2	2	2	2	2	2	2	1
1	2	3	3	3	3	3	3	2	1
1	2	2	2	2	2	2	2	2	1
1	1	1	1	1	1	1	1	1	1

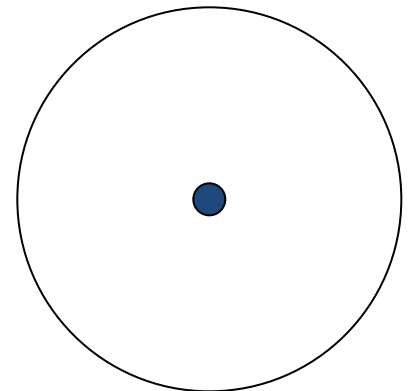
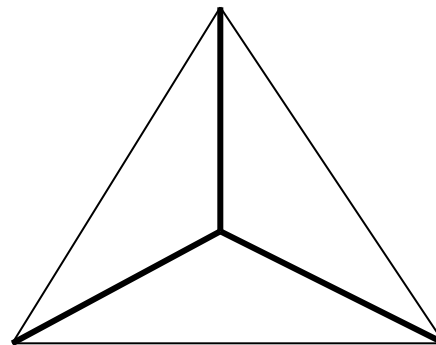
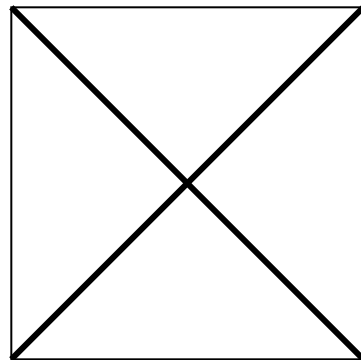
- Centers of Maximal Disks
That touch boundary in
two points
- Connectivity not preserved



Medial Axis - Grassfire

- Suppose that a fire line propagates with constant speed from the contour of a connected object towards its inside, then all those points lying in positions where at least two wave fronts of the fire line meet during the propagation will constitute a form of a **skeleton**

- Examples



Medial Axis Transform

